

Managing Skills in Organizations: Evidence from a Field Experiment*

We investigate the dynamics of skill management in organizations through a natural field experiment with 2,582 service technicians. Employees in both treatment and control groups could self-assess skills and propose training measures to managers. An intervention restricting managers' access to objective skill information reduced training intensity, work performance, and job satisfaction – indicating underinvestment in human capital, even with employee-initiated development. Using detailed personnel records and survey data, we show that the intervention dampened both employees' efforts to identify training needs and managers' engagement in skill management. In particular, high-skill employees received less training aimed at broadening their skill set, and performance declined due to longer completion times for complex assignments. These findings highlight the pivotal role managers play in fostering skill development and workforce productivity.

Keywords: Skill Management, Monitoring, Training, Field Experiment
JEL Codes: J24, J28, M12, M53

*One of the authors received an internship contract from the firm to facilitate access to confidential data. The firm has exerted no influence on the data analysis or scientific conclusions drawn in this paper. This study has been pre-registered on OSF and was approved by an IRB.

1 Introduction

Human capital is a critical driver of firm performance, which is why firms have a vested interest in a highly-skilled workforce (Bartel, 1989; Black & Lynch, 1996; Bloom, Sadun, & Van Reenen, 2010). To benefit from technological innovation, firms must ensure that their employees are prepared for the ever-evolving work requirements (Acemoglu & Restrepo, 2022; Autor, Chin, Salomons, & Seegmiller, 2024; Brynjolfsson, Mitchell, & Rock, 2018). As acquiring new skills is central to this transformation, understanding and developing employee skills has become crucial for firm success. Historically, measuring and monitoring employee skills has been challenging and costly, but the widespread adoption of learning technologies has significantly lowered the barriers to assessing employee skills and to quantify the investments in developing them.¹ However, there is little causal evidence on the role of managers in this process.

In this paper, we study how the availability of such information to managers affects the skill management process. Prior evidence from the field suggests that providing managers with information about employee behavior generally increases performance in the target dimension, as it reduces moral hazard and facilitates targeted interventions by the manager.² However, this logic is less straightforward when the data concern employees' skills rather than their realized performance. On the one hand, employees themselves benefit from stronger human capital and thus may have an incentive to learn about their skills and improve them out of their own volition. Moreover, skill levels are highly relevant to one's self-image and, consequently, to how others - especially superiors - perceive them. It has often been claimed that reputational concerns are key drivers of effort in organizations, or as put by Holmström (2017) in his Nobel

¹Despite the importance of skills as a research topic, the literature shows little consensus on a definition of employee skills. In this paper, we focus on technical skills, which are "combinations of cognitive and non-cognitive skills used to accomplish specific tasks" (Brunello & Wruuck, 2021, p. 1146).

²For instance, prior research shows that moral hazard explains why auto mechanics are reluctant to use checklists to monitor their performance (Jackson & Schneider, 2015), giving managers in a retail bank access to objective performance measures increases profits (Manthei & Sliwka, 2019), randomly installing GPS tracking devices in a Nigerian trucking firm increases average speed (Rochambeau, 2019), increasing the visibility of the monitoring technology improves task performance in some dimensions (Jensen, Lyons, Chebelyon, Bras, & Gomes, 2020), and introducing performance review meetings in a retail chain raises profits (Manthei, Sliwka, & Vogelsang, 2023).

Prize lecture: “The craving for appreciation and the desire to impress superiors explains why a mere change in the accounting system can have a big impact on the behavior of employees” (p. 1772). Managerial access to information on employee skills may therefore generate two opposing forces driven by reputational concerns: It may encourage employees to invest in skill development because such investments allow them to demonstrate and receive praise for high levels of skill (Bénabou & Tirole, 2006; Ewers & Zimmermann, 2015; Falk & Szech, 2016). Yet it may also reduce employees’ willingness to assess their own skills if they anticipate that lower skill levels will be exposed and, in turn, negatively affect their reputation and career prospects (Baader, Bowen, Hochleitner, & Mills, 2024; Bursztyn, Egorov, & Jensen, 2019; Heursen, Friess, & Chugunova, 2023).

We leverage a unique empirical setting to study a skill management process in detail, utilizing a rich data set that spans skill assessments, training participation, supervisor interactions, productivity metrics, and survey results. In a natural field experiment involving 2,582 service technicians at a large technology company, we vary managers’ ability to monitor employee skills. Employees are regularly invited to complete online assessments regarding job-specific technical skills. Managers are instructed to hold quarterly development meetings to discuss training needs. Prior to the intervention and within the control group, employees were encouraged to share their skill assessment outcomes with managers in these meetings. In the treatment group, however, employees were instructed to withhold these results, and managers were told not to request them. As employees in both groups continue to be encouraged to recommend training measures, the experiment effectively restricts managerial access to hard skill data while preserving employee initiative in identifying training needs.

We illustrate the tension between information use and image concerns in a simple formal framework, predicting that while monitoring should boost engagement for high-skill employees by providing both better-targeted training and positive reputational payoffs, its effect on low-skill employees is ambiguous. For the latter group, the anticipated embarrassment of exposing

knowledge gaps can create a counter-force that deters participation. Our field experiment thus tests these predictions by restricting manager access to assessment data, exploring whether removing monitoring can mitigate reputational fears and ultimately encourage greater skill investment among low-performing employees.

Our key results are the following: Reducing managers' ability to monitor employee skills lowered the overall training intensity substantially as the treatment reduced the number of trainings booked by about 50%. It also led to a decline in employee performance by causing significantly longer task execution times. A back-of-the-envelope calculation indicates that this decline in efficiency would translate into approximately USD 1,040,000 in additional annual personnel costs. Furthermore, contrary to the common view that monitoring diminishes psychological well-being, our results demonstrate that its removal did not improve job satisfaction but, in fact, led to a significant decline.

To explore the underlying mechanisms in more detail, we examine the roles of employees and managers in the skill management process. We begin by documenting evidence in line with a complementarity between the actions of both parties: training intensity is highest when employees assess their skill level to identify training needs, *and* managers take time to discuss those needs. Conversely, training intensity is substantially lower when either party does not contribute to the development process. We then demonstrate that the treatment reduced the contributions from employees and managers to this process. On the one hand, employees were less likely to participate in the skill assessments – and in contrast to our hypothesis – especially those with lower prior skill levels. At the same time, managers reduced their engagement by conducting development meetings less often, in particular for high-skill employees. Survey evidence further shows that, in line with the role of image concerns, the stated motivation of high-skill employees to participate in skill assessments to demonstrate their competence was reduced by the treatment. However, we do not find evidence for an encouragement effect of the treatment on the low-skill employees due to a reduction in the fear of being blamed.

Although a decline in human capital investments among already high-skill employees may not appear problematic at first glance, we show that the intervention undermined the acquisition of *new* skills among these employees. In line with this observation, we find that productivity losses are driven by longer completion times for complex tasks, which require an advanced skill set. Taken together, our findings underscore the vital role of skill monitoring by supervisors – not only in mitigating skill gaps among low-skill employees, but also in sustaining continuous learning and up-skilling among the high-skilled.

Our paper contributes to the study of training within firms. While there is a growing literature that quantifies the direct and indirect returns of training in organizations ([Adhvaryu, Nyshadham, & Tamayo, 2023](#); [De Grip & Sauermann, 2012](#); [Espinosa & Stanton, 2023](#)), “little work has covered how firms decide on training investments” ([Hoffman & Stanton, 2024, p. 78](#)). Prior research shows that middle managers play an important role in employee development and facilitating the managerial assignment of training through additional decision rights and performance information improves employee performance ([Adhvaryu, Nyshadham, & Tamayo, 2024](#); [Diaz, Nazarrett, Ramirez, Sadun, & Tamayo, 2025](#); [Friebel, Heinz, & Zubanov, 2022](#)). However, current papers suggest that the standard rationale behind the managerial assignment of training is to focus on low- or under-performing employees with the goal of raising performance up to that of the average employee ([Delfino, Garnero, Inferrera, Leonardi, & Sadun, 2026](#); [Diaz et al., 2025](#); [Sandvik, Saouma, Seegert, & Stanton, 2026](#)). In contrast to this approach, we find that investing more time and effort in broadening the skill set of *high-skill* employees can be more fruitful in an environment that is characterized by frequent technological change. Our paper further contributes to the research on management practices ([Bloom, Sadun, & Van Reenen, 2007, 2012](#)). Several studies have shown that employee monitoring is positively related to firm-level productivity and profitability ([Gosnell, List, & Metcalfe, 2020](#); [Jensen et al., 2020](#); [Rochambeau, 2019](#)), as it solves classical moral hazard problems ([Holmström, 1979](#); [Merchant & Van der Stede, 2017](#)). While most research in economics documents positive effects

of monitoring on performance, psychological research suggests that increased levels of monitoring can have a negative effect on employee well-being (Deci, Connell, & Ryan, 1989; Nebeker & Tatum, 1993; Ravid, White, Tomczak, Miles, & Behrend, 2023) – a point often brought up by the popular press and also in our discussions with the firm (Kantor & Sundaram, 2022; Murty & Karanth, 2022; Shrikant, 2023). Therefore, one might expect that removing managers' ability to observe the outcomes of the skill assessment would reduce the perceived pressure and lead to higher job satisfaction. We show, however, that the monitoring of skills can be beneficial for both productivity and job satisfaction. Finally, our results also contribute to the literature on career and image concerns in organizations (Bénabou & Tirole, 2006; Holmström, 1982) by unpacking two specific reputational drivers. Prior research shows that in situations of managerial observation, people strategically adjust their behavior to make a good impression (Burks, Carpenter, Goette, & Rustichini, 2013; De Janvry, He, Sadoulet, Wang, & Zhang, 2023; Ewers & Zimmermann, 2015) or to avoid a negative impression (Baader et al., 2024; Heursen et al., 2023). Our results are in line with the former motive, as high-skill employees reduce their effort in the knowledge test when it is no longer observable by the manager. We find no evidence for the second motive, according to which employees avoid participation due to poor expected performance.

The paper proceeds as follows: Section 2 illustrates the operating environment of the firm and its skill management process. Section 3 lays out the experimental manipulation and shows descriptive statistics of key dependent variables. Section 4 outlines the conceptual framework and derives predictions for the analysis. In Section 5 we show causal effects of the intervention on training, work performance, and employee well-being. We then analyze potential mechanisms in Section 6 to understand the role of managers and employees in the skill management process. In Section 7 we show how these mechanisms affect training and work performance. Section 8 concludes.

2 Institutional Setting

We run the experiment in the technical service division of a large service organization. The organization employs service technicians operating in 249 teams, 125 of which are part of this experiment. Service technicians mostly work independently and their primary responsibility is to install and maintain the firm's products and services. While some assignments can be performed remotely, most of the work is done by one technician at the client site, i.e., in private homes or business offices. Within each team, every employee is responsible for a designated geographical area. Employees receive their daily work schedules from a regional dispatcher. The team manager coordinates the content of the work schedule with the dispatcher and ensures that the technicians on their team have the skills required to complete their assignments within the allotted time frame (see Figure A1 for an overview of the relationship between employee, manager and regional dispatcher). Given the technical nature of the work tasks and ongoing technological changes in the industry, continuous skill management is essential to maintaining high levels of productivity and customer satisfaction. An overview of the firm's skill management process is presented in Figure 1. The first step in this process involves employees participating in regular knowledge assessments to evaluate their skills. Assessments are conducted twice every year for each of six core technical skill domains (hereinafter referred to as "skills"). Each employee has a skill profile specifying a subset of the six core skills, and these skill profiles determine the work assignments that can be allocated to them. Most employees have two or three skills and they are automatically invited to the respective assessments. Hence, a typical employee is supposed to take part in around 5 skill assessments per year. Managers may also assign new skills to employees in order to broaden their skill sets. Skill assessments are administered through multiple-choice questions on an online platform, accessible via tablet, phone, or laptop. Employees receive an automated invitation and are given a designated one-hour time frame during which no work assignments are scheduled. Each assessment contains 20 questions, randomly selected from a pool of 60 questions per

skill. If an employee answers a question incorrectly, the correct answer, along with a brief explanation, is provided before they proceed to the next question. All questions are created by experienced technicians and reviewed by the works council to ensure that they cover the actual work content appropriately.

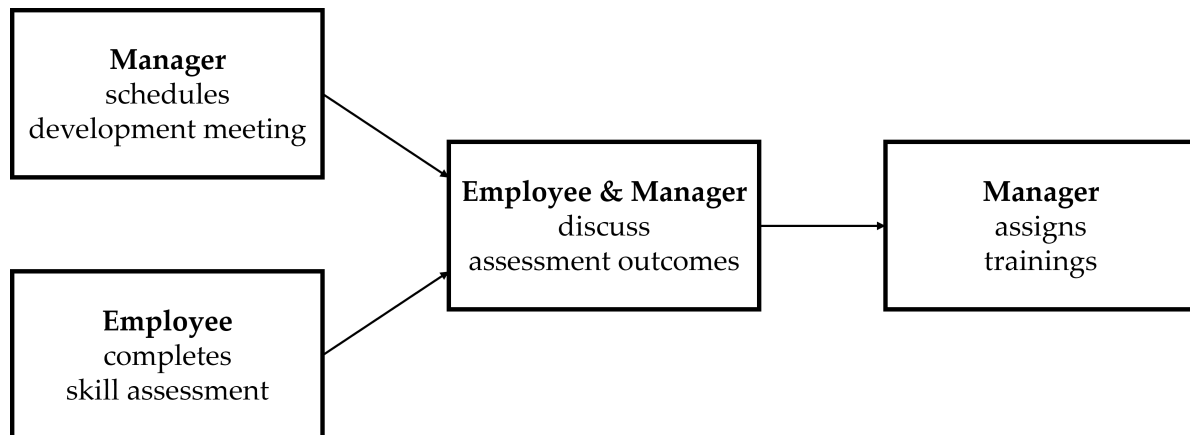


Figure 1: Overview of the skill management process of the firm.

After completing the assessment, employees receive an overall rating of their current skill level, along with a detailed record of their responses to every question. They can then choose from a set of suggested training measures, which they may recommend directly to their manager. During this step, employees can also write a personal message that will be transmitted along with their suggestions (see Figure A2 for screenshots). In the following step, managers and employees meet to discuss the results of the skill assessment and the employees' training proposals. Training usually serves one of two purposes: Training should either fill gaps in skills employees already have ("skill preservation") or broaden their skill set by introducing them to a new technology or exposing them to more advanced, non-routine problems ("new skill acquisition"). It is thus possible that two employees receive training on the same broader topic, but the content differs because the training has a different purpose. Given the technical nature of the work, training mostly takes place in person. In these training sessions, employees practice work on real products with their daily tools under the supervision of an instructor.

Before the intervention and in the control group, employees are asked to bring their assessment results to the development meeting. Managers then decide on the allocation and scheduling of training sessions, based on the employees' proposals and the results of the skill assessment. Since the introduction of the learning platform, the firm's management and its works council have been debating which information needs to be available to the manager.³ The works council argues that showing rather detailed assessment results to managers is excessive and potentially degrading, suggesting that employees might be discouraged from participating in assessments due to fear of being perceived as incompetent after a poor performance in a skill assessment. To gain a deeper understanding of these concerns, we conducted ten short phone interviews with five managers and five employees from different teams, who were referred to us by the firm. In these interviews, some employees indeed expressed that fear of being blamed for poor outcomes might contribute to a reluctance to participate in skill assessments or to share their results in the development meeting. Although the firm's leadership has historically been in favor of monitoring employee outcomes, they were open to reducing it if doing so would improve assessment participation and encourage training, particularly among low-skill employees.

3 Experimental Setup

We randomly assigned 125 teams ($N = 2,582$ employees) to either a treatment or the control group.⁴ At the start of the intervention, all employees were informed about upcoming changes to the learning platform via email and directly through the platform when completing a skill

³In Germany, employees have a right to set up an employee-elected works council in establishments with more than 5 employees. Firms need the consent of the works council when implementing policies to monitor employees' performance. § 87 (1) of the [Works Constitution Act](#) [BetrVG], for instance, specifies that "The works council has a right of co-determination [...] in the introduction and use of technical devices designed to monitor the behavior or performance of the employees".

⁴All 249 operational teams were initially included in the randomization process and allocated to one of four different groups, two of which are the focus of the present study. In another pre-registered experiment (pre-registered on OSF under [ID JXAE2](#)), we compare the treatment group in the present experiment to the remaining two experimental groups. As each group receives different features, we cannot use data from the other two experimental groups. The communication and data collection processes were identical for all 249 teams.

assessment (see Figure A2). All managers and employees were also invited to participate in virtual Q&A sessions to better understand the changes in their respective group and role. In both groups, we introduced a new feature of the learning platform through which employees were encouraged to propose potential training measures to their managers. In the treatment group, employees were explicitly instructed *not* to share their assessment outcomes with their managers, and managers were told not to request them, but rather to focus on the training measures proposed by their employees (Figure A2 shows an English translation of the verbatim communication to the employees). Since managers receive hard information about individual employee performance only in exceptional cases, the skill assessments were one of the few sources of hard information on employees' training needs.

Even though we instructed both managers and employees not to discuss the results of their skill assessments and rather focus on training allocation, we cannot assure full compliance with the treatment. Hence, we note that the treatment effect estimates we report below likely tend to underestimate the true effects that would be obtained with perfect compliance. While we cannot assess treatment compliance directly, we have access to data from a new messaging feature, which was implemented for all employees and which employees could use to send messages to their supervisor within the skill management platform. Analyzing the messages sent through the new feature, we find no explicit evidence for non-compliance in these messages – for instance no employee sent their assessment outcome to their manager. However, some employees indicated that they perceived their own assessment performance as sub-par, potentially to justify further training.

The firm is organized in six departments that comprise a total of 37 regions. Treatment assignment was stratified at the regional level. We further included team size, average team assessment participation and assessment scores as stratification variables. The experiment was conducted from July 1, 2022, to March 31, 2023. The post-experimental survey was distributed via email on behalf of the CEO in January 2023. All participants who completed the survey

received a compensation of €7 via bank transfer. This particular survey was conducted by the researchers and respondents were informed that the company would only receive aggregated responses and anonymity would be preserved. Compensation for survey participation was paid from university research funding. Unless stated otherwise, we use all available data from the beginning of 2022 until the end of the post-experimental survey in March 2023. For the data analysis, we excluded employees who switched treatment conditions before the end of March 2023, as they may have been part of different treatment groups.⁵ Table 1 provides summary statistics for relevant pre-experimental outcomes for the treatment and control group. In addition to the stratification variables – assessment participation, standardized assessment score, and team size – we also consider key dependent variables for the analysis, such as the number of trainings per capita and the average duration of key work tasks per team.⁶ Prior to the intervention, employees on average took part in about 80% of the skill assessments relevant to them and attended about 0.5 trainings per year.

⁵In Table A16, we re-run the main analyses of the paper excluding all teams in which at least one member has been exposed to both treatments before the end of the post-experimental survey to avoid violations of the stable unit treatment value assumption (SUTVA). This leads to the exclusion of 4 teams (3 control teams and 1 treatment team), but has no meaningful effect on the results of the analyses.

⁶Note that it was not possible to measure the number of trainings and development meetings on an individual level prior to the experiment. Instead, data collection for this time period was limited to the team level.

Table 1: Balance Table

	Control (N=63)		Treatment (N=62)		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Assessment Participation	0.81	0.16	0.80	0.17	−0.02	0.54
Assessment Score	0.00	0.28	0.00	0.34	0.00	0.95
Number of Trainings	0.25	0.30	0.24	0.30	−0.01	0.79
Supervisor Meetings	0.86	0.34	0.78	0.41	−0.07	0.35
Troubleshooting (in min.)	34.41	3.79	34.05	4.14	−0.36	0.62
Customer Service (in min.)	7.19	1.27	6.68	0.98	−0.51	0.02
Quality Control (in min.)	10.34	2.08	10.54	2.28	0.20	0.63
Documentation (in min.)	4.87	0.36	5.02	0.48	0.15	0.06
Team Size	20.00	2.16	20.13	2.80	0.13	0.77

4 Conceptual Framework

Investing in employee skills may require coordinated effort from two parties: the employee and the manager. To fix ideas, consider the following framework, developed formally in the appendix, which captures this joint investment problem and its dependence on whether managers can monitor employees' skill levels.

The basic investment problem. Consider an organization where employees need to develop their skills continuously. Employees can invest time and effort to identify their own skill gaps (in our setting, by taking part in an online knowledge assessment), while managers must invest effort to engage in a structured development conversation (in our setting, by conducting a quarterly development meeting) and schedule training. Both managers and employees benefit from effective skill development when it results in higher levels of human capital and higher productivity. But investments are costly for each party in terms of time and attention, and their opportunity costs vary across individuals and situations. Crucially, the two investments are *strategic complements*: each party's effort becomes more valuable when the other party also

contributes. An employee who participates in the assessment but whose manager does not engage in the conversation gains less; and a manager who prepares for a meeting with an employee who has not identified any training needs has little to work with. The highest human capital gains are thus realized when both parties invest, and the gains are greater the more intensively both participate.

The role of monitoring. The central question in the framework is whether giving the manager access to the employee's assessment results – the monitoring technology – increases or decreases the overall level of investment by both parties. On one hand, monitoring makes the manager's effort more productive: when she can see exactly where an employee's knowledge gaps lie, targeted training recommendations become more useful, which also raises the manager's incentive to engage. Through strategic complementarity, this higher manager engagement comes along with higher employee's own incentive to participate in the assessment. On the other hand, monitoring introduces a social and reputational dimension to the interaction: when the employee knows that the manager will see the results, participating in the assessment is no longer a purely private exercise. For employees who expect to perform well the development meeting becomes an occasion to demonstrate competence and receive recognition, which adds an additional incentive to participate. For employees who expect to perform poorly, however, visibility can create the opposite force: the anticipated embarrassment of discussing a weak result with one's manager may outweigh the human capital gains from participating, discouraging assessment participation.

Key predictions. The framework yields two predictions that shape the empirical analysis. First, for high-skill employees – those who expect to perform well – monitoring is unambiguously beneficial. It simultaneously increases the human capital returns to the meeting (through better targeting of training) and adds a positive reputational payoff to the interaction. Both forces push employee participation and manager engagement in the same direction, reinforcing each other through strategic complementarity. Second, for low-skill employees – those who

expect to perform poorly – the effect of monitoring is ambiguous. Whether monitoring raises or lowers investment depends on the magnitude of reputational concerns: if the fear of embarrassment is sufficiently small relative to the expected human capital gains, monitoring still benefits both parties; but if reputational concerns and the fear of blame are strong enough, monitoring may deter assessment participation among low-skill employees and, through complementarity, also reduce manager engagement for this group.

As described above, the treatment in our experimental setting removes managers' access to assessment outcomes: employees are instructed not to share their results, and managers are told to focus only on the training proposals put forward by the employee. The framework predicts that this removal of monitoring should always dampen training investments for high-skill employees. However, it also predicts that – if the fear of embarrassment of low performers is sufficiently strong – it may encourage low-skill employees to participate more frequently, as the fear of being blamed for low skills is reduced.

5 Results

We organize the empirical analysis in three steps. We begin with training, which is the most tangible measure of organizational investment in employee skills, and then examine whether the intervention also affected work performance and employee perceptions such as job satisfaction. Throughout the paper, we use individual data whenever available. This includes all results on employee skills and training, as well as the results from the post-experimental survey. Due to regulations from the works council of the firm, employee performance data is aggregated on the team level. Therefore, all analyses on employee performance, such as assignment completion times and quality are conducted on the team level. Data levels are indicated in each table below the estimation results.

5.1 Trainings

To begin with, we examine the effect of the treatment on employee development, as measured by the number of trainings assigned by the manager. The average treatment effect on the number of trainings is depicted in Figure 2 and the corresponding regression results are displayed in Table 2, where we also analyze the likelihood of an employee receiving at least one training in the treatment period. In columns (2) and (4), we control for pre-treatment participation in the skill assessment and the number of skills prior to the experiment. As Figure 2 shows, the number of trainings per employee in the first 6 months of the treatment period is reduced by 0.19 compared to the control group average of 0.37, which corresponds to a 51.89% reduction ($p = 0.036$). Estimating the extensive margin effect – i.e. the likelihood of receiving any training at all – we find that employees in the treatment group are 7.23 percentage points less likely to receive any training in the first 6 months of the treatment period, which translates to a 46.85% reduction ($p = 0.03$) in comparison to the control group (see columns 3 and 4 of Table 2). Therefore, reducing managers' ability to directly monitor and manage employee skills substantially reduces training intensity and the overall likelihood of receiving training.⁷ Overall, these results show that restricting managers' access to assessment results and participation information substantially reduces the investment in employee development. The next question is whether this reduction merely changes the amount of training provided or is accompanied by economically meaningful consequences for the organization. We therefore turn to employees' actual performance.

⁷In Table A1, we replicate our results using a Pseudo-Poisson ML estimator for the treatment effects in columns (1) and (2) of Table 2 and a logit model to estimate the treatment effects depicted in columns (3) and (4). We further re-estimate the treatment effect on the number of trainings on the team level where we can control for pre-treatment values and target a longer time period (12 months instead of 6), which yields similar results.

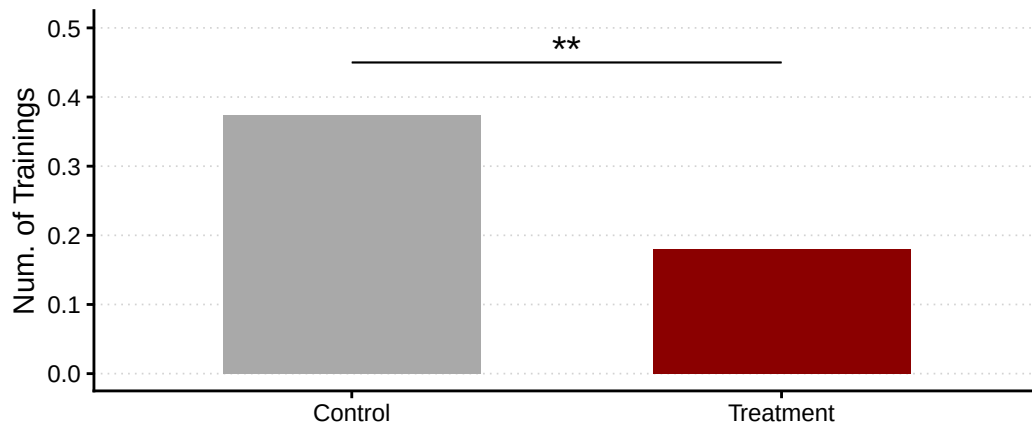


Figure 2: Treatment Effect on the Number of Trainings per Employee.

Table 2: Treatment Effects on Trainings

	Num. of Trainings		Prob. of Training	
	(1)	(2)	(3)	(4)
Treatment	−0.194** (0.091)	−0.213** (0.097)	−0.072** (0.033)	−0.077** (0.035)
Prior Participation		0.268*** (0.070)		0.109*** (0.025)
Number of Skills		0.021 (0.031)		0.012 (0.010)
Constant	0.374*** (0.077)	0.121 (0.119)	0.154*** (0.027)	0.042 (0.044)
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	125	125
Observations	2582	2234	2582	2234
R ²	0.011	0.022	0.013	0.025

Note: Table shows estimates from linear regressions on the number of assigned trainings per employee and the likelihood of receiving at least one training assignment. Columns (1) and (2) use the number of trainings as a dependent variable, which is the sum of all trainings associated with key skills. In columns (3) and (4) the dependent variable is a binary indicator for receiving training which takes value 1 if an employee has received at least one training and 0 otherwise. Prior participation captures the participation rate in the skill assessment prior to the experiment and the number of skills measured at the same time point. Standard errors are clustered at the team level and reported in parentheses. **, *** indicate significance at the 5% and 1% levels, respectively.

5.2 Work Performance

If restricting skill monitoring weakens employee development, its operational consequences should be most visible in work activities for which employee knowledge and problem-solving ability matter. We therefore examine completion times across the four most important tasks of the technicians' assignments. To understand the effects of the intervention on employee performance in detail, we utilize the firm's performance tracking system, which measures the time a technician takes to complete a task in a given work assignment. We focus on the four work phases that are common for almost every work assignment and account for 95% of all observations in the data set.⁸ The variable Customer Service ($M = 6.27$ min) includes all tasks associated with customer interaction before and during the appointment. For most clients, employees are required to call the customer to understand their infrastructure and their current issue. This short conversation often serves as a precursor to on-site troubleshooting. Troubleshooting ($M = 43.29$ min) is the core of a technician's job and involves all tasks associated with problem-solving at the client site. Troubleshooting is the most time-intensive task for which technical knowledge and training are most likely to play a significant role. Quality control ($M = 10.22$ min) follows troubleshooting and involves assessing the client's infrastructure, typically after resolving the issue. If any further problems remain, this serves as documentation for a follow-up visit. Finally, documentation involves all tasks associated with paperwork or data entry ($M = 4.48$ min). As there is a slight imbalance in prior task times (see Table 1), we perform difference-in-differences regressions for the average time needed to complete each step in the work assignment. We include team fixed effects and account for assignment-specific time trends to account for seasonal changes in product sales (e.g. holiday seasons).⁹ As Table 3 shows, reducing access to employee skill assessments reduces employee performance, as is

⁸One row in the data set represents the average time to complete a specific activity in a given week across all members of one team. The tasks presented in Table 3 cover 95% of all rows in the data set. The remaining tasks concern, for instance, the delivery of equipment to a work site or manual programming.

⁹A detailed analysis regarding the nature of the work assignment follows in Section 7, where we further differentiate work assignments by their degree of complexity.

evident from the increased average assignment times in two key phases. Column (1) shows that teams in the treatment group need 0.51 minutes longer to complete the same customer service assignments as teams in the control group. In comparison to the control group mean prior to the experiment, the treatment thus caused an 8.11% increase in the average time spent on customer service tasks ($p = 0.046$). The average time spent on troubleshooting activities increases by 1.38 minutes (see column 2) or 3.19% ($p = 0.074$). We do not find effects on the last two steps of the work assignment, such as quality control (column 3) and documentation (column 4). As reported in Table A2 in the Appendix, we find no evidence that these higher task execution times came along with a higher customer satisfaction or reduced the share of assignments that required a follow-up visit.¹⁰ A back-of-the-envelope calculation illustrates that the personnel costs associated with the treatment can be substantial. Even if employees complete only two in-person customer visits per day, the estimated time increase of the treatment is equal to 33500 hours per year. This corresponds to the annual working hours of 20 full-time employees or about USD 1 million in estimated personnel costs.¹¹

¹⁰Figure A6 and Figure A7 plot the corresponding event study estimates for the treatment effects on customer service and troubleshooting tasks by quarter (see also Table A8 where we replicate the results from Table 3 at the quarter level).

¹¹In this calculation, we neither consider the time saved by the reduction in training, nor a potential relative decrease in wages that stems from the reduction in skill acquisition. Therefore, our calculations should be considered as an upper bound. Wage information was retrieved from [the German Union for Technical Workers \(IG Metall\)](#).

Table 3: Treatment effects on task completion times

	Customer Service	Troubleshoot- ing	Quality Control	Documenta- tion
	(1)	(2)	(3)	(4)
Treatment	0.509** (0.252)	1.381* (0.766)	−0.051 (0.462)	−0.033 (0.100)
Team FE	Yes	Yes	Yes	Yes
Product × Week FE	Yes	Yes	Yes	Yes
Data Level	Team	Team	Team	Team
DV mean	6.27	43.29	10.22	4.48
Clusters	121	121	121	121
Observations	33 835	42 801	34 700	38 140
R ²	0.087	0.268	0.292	0.171

Note: Table shows average treatment effects from difference-in-differences regressions on the average time per task in minutes. All models include team as well as Product × Week fixed effects. Assignment types are usually associated with a specific technology, but do vary with respect to their degree of complexity. ‘DV mean’ depicts pre-experimental averages from the control group. Standard errors are clustered at the team level and reported in parentheses. *, ** indicate significance at the 10% and 5% levels, respectively.

5.3 Employee Perceptions

To study the effect of the treatment on job satisfaction, we use data from the post-experimental survey, which can be linked with individual skill assessment and training data (see Section A.5 for all survey items). The overall participation in the post-experimental survey is 36.76%. While survey participation was higher in the control group, participants in both groups exhibit very similar characteristics with respect to prior participation in skill assessments and achieved assessment scores (see Table A6).¹² Analyzing treatment effects on psychological outcomes

¹²Even though the selection into the survey does not appear to be driven by observable pre-treatment characteristics, unobserved differences might still lead to biased results. To investigate this potential issue in more detail, we calculate generalized Lee bounds (Semenova, 2025) to obtain a more robust estimate of the causal effect of the treatment on the survey outcomes in Table 4 in the presence of selection bias. As Table A17 shows, the estimated identification intervals have the same sign as the conventional estimates, which provides no evidence that differential survey participation reverses the direction of the results.

shows a universally negative impact: As shown in Table 4, the treatment significantly reduced job satisfaction (column 1, survey item “*All things considered, I am quite satisfied with my current job.*”) and supervisor support (column 2), which refers to the perceived attention and effort managers invest in their employees’ development (seven items, e.g. “*My manager is interested in what I’m currently learning.*”). It is important to recall that in the control group, employees had the same access to the skill assessments and were also encouraged to share their training needs with their managers. Nevertheless, the monitoring of skill levels by supervisors thus fostered rather than undermined job satisfaction, a finding we explore in more detail below.

Table 4: Treatment effects on employee perceptions

	Job Satisfaction	Supervisor Support	Turnover Intention
	(1)	(2)	(3)
Treatment	−0.200** (0.077)	−0.214** (0.097)	0.088 (0.079)
Constant	0.094* (0.053)	0.098 (0.062)	−0.040 (0.055)
Data Level	Individual	Individual	Individual
Clusters	125	125	125
Observations	775	848	819
R ²	0.010	0.011	0.002

Note: Table shows estimates from linear regressions on standardized measures of job satisfaction, supervisor support and turnover intentions as reported by the employees in the post-experimental survey. Job satisfaction and turnover intentions are measured using single-item measures from the Linked Personnel Panel (Kampkoetter et al., 2016), our measure of supervisor support is based on a four-item survey from Decius, Schaper, and Seifert (2021) and a three-item scale from Rimann and Udris (1999). All questions are answered on a 7-point Likert scale from 1 = ‘strongly disagree’ to 7 = ‘strongly agree’. Standard errors are clustered at the team level and reported in parentheses. *, ** indicate significance at the 10% and 5% levels, respectively.

6 Mechanisms

Having established that reducing managers' ability to actively manage employee skills reduces training intensity, performance, and job satisfaction, we now investigate the underlying mechanisms in more detail. As illustrated in Section 4, successful skill management requires both the employees and their respective managers to invest time and effort: Employees need to be willing to identify skill gaps by taking part in the skill assessment, and managers need to schedule development meetings to discuss assessment results and assign training. Indeed, a simple descriptive regression of training intensity on assessment participation, meeting incidence, as well as their interaction shows a complementary relationship between both contributions (see column 2 in Table A3 in the Appendix). That is, training intensity is low when either of the two contributions is absent. In the following, we study the contributions of both sides in turn. We first consider the employee perspective, where we analyze treatment effects on participation and results in the skill assessments. Then, we study the manager's perspective where we investigate the role of development meetings for skill development.

6.1 Employee Investment

To study treatment effects on employee actions, we focus on the three most important skill assessments that correspond to the majority of work assignments of the firm and for which both pre- and post-experimental data are available. Figure A3 shows that these skills cover 89% of all work assignments.¹³ Employees are only invited to a skill assessment if they have been assigned to the corresponding skill. The key outcome variable that we consider here is thus average participation for assessments associated with these three skills. Prior to the experiment, the average share of correctly answered questions was 53.03% in the most widespread skill assessment. In the second and third most widespread skill assessments, this share was 64.02%,

¹³In Table A4, we show treatment effects for all skill assessments in the treatment period across all skills. While it is possible to re-take the assessment, we find only six instances in our data set where an employee participates twice in the same assessment. Therefore, we only consider assessment scores in the first try for every assessment.

and 71.64%, respectively. Our final data set includes 19,447 assessment results, which we standardize for each assessment separately and then create the mean standardized score across all three assessments. Table 5 shows treatment effects on assessment participation and the average standardized score. As column (1) shows, the treatment reduced the average participation rate by 6.59 percentage points from 79.03% to 72.43% ($p < 0.001$). Hence, rather than encouraging employees to take part in the assessment to learn about their own skill gaps when they can do so in private, the intervention reduced the likelihood that employees assess their own skills. We find no evidence, however, that the treatment affected the average assessment result of those who took part in the skill assessment.

It is conceivable that apart from the reduction in monitoring, the treatment may have sent a signal to the employees that skill development – and therefore the participation in the assessment – is no longer a priority of the firm. To investigate this conjecture more explicitly, the analyses in Table A9 investigate whether treated employees felt less obliged to take part in the assessment because it is part of the job (*“I take part in the assessment, because it is part of my job.”*) or because it was explicitly demanded by their managers (*“...because my manager expects me to do so.”*). The results in Table A9 show no significant effect of the treatment on any of these items. It thus appears that employees are still aware that they are expected to complete the skill assessment, but potentially decided to avoid it now that non-participation is not directly observable for the manager anymore.

A key conjecture presented in Section 4 was that reducing managers’ ability to observe assessment outcomes affects employees differently depending on their prior skill level. If employees with *lower* skill levels are discouraged from performing assessments due to fear of negative judgement, the treatment may have positively impacted participation by ensuring results remain private. Employees with *higher* skill levels on the other hand may have reduced assessment participation, as they can no longer receive additional recognition for their excellence. To investigate this, we classify those employees that had taken part in at least

one of the focal skill assessments prior to the intervention and achieved an above-median score as *high-skill*, and the remainder as *low-skill*.¹⁴ The corresponding results are shown in Table 5. Contrary to our expectations, column (2) shows that the reduction in assessment participation seems to be driven by low-skill employees ($p = 0.001$). Hence, the intervention did *not* encourage this group to take part in the assessment. On the other hand, participation among high-skill employees remains mostly unaffected ($p = 0.771$). If anything, we find weak evidence that the treatment reduced assessment scores of high-skill employees relative to those with lower skills ($p = 0.071$).¹⁵

¹⁴Note that prior non-participants are thus also classified as *low-skill*. In Table A5, we show that there is a strong correlation between test participation and test outcomes, i.e., past non-participants have substantially lower test outcomes when taking part in a test now and current test participation is higher for employees with better past test outcomes. Additionally, Figure A5 shows that prior non-participants on average have fewer skills compared to any other group and it is therefore unlikely that these prior non-participants are in fact high-skill employees who were not motivated enough to participate in the skill assessments.

¹⁵In Figure A4, we investigate more specific subgroups comparing test participation and test scores between treatment and control group for the prior non-participants and by quartile of prior assessment scores. Notably, the reduction in test participation is driven by those employees who had not taken part in prior tests, while high-skill employees seem - at least to a certain extent - to reduce their assessment scores in response to the treatment.

Table 5: Treatment Effects on Assessment Participation and Assessment Scores

	Assessment Participation		Assessment Scores	
	(1)	(2)	(3)	(4)
Treatment	−0.066*** (0.022)	−0.093*** (0.028)	−0.015 (0.050)	0.044 (0.063)
Treatment × High-Skill		0.099*** (0.027)		−0.134* (0.073)
High-Skill		0.058*** (0.017)		0.837*** (0.052)
Constant	0.790*** (0.015)	0.761*** (0.018)	−0.054 (0.034)	−0.440*** (0.042)
Wald-Test High-Skill		0.771		0.109
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	124	124
Observations	2321	2232	2210	1988
R ²	0.009	0.035	0.000	0.196

Note: Table shows estimates from linear regressions of assessment participation (columns 1-2) and assessment scores (columns 3-4) on a treatment indicator. Participation denotes the number of assignments an employee has taken part in during the treatment period, divided by the number of assignments they have been assigned to. Assessment scores depict the average standardized ratio of correctly answered questions on the knowledge assessment. Columns 1 and 3 show results for the full sample. Columns 2 and 4 report interaction models using the variable 'High-Skill', which takes value 1 for employees with an above-median average assessment score prior to the experiment and 0 otherwise. 'Wald-Test High-Skill' reports p-values from a Wald test of the null hypothesis that the treatment effect for high-skill employees is equal to zero (Treatment + Treatment × High-Skill = 0). Standard errors are clustered at the team level and shown in parentheses. *, **, *** indicate significance at the 10%, 5%, and 1% level, respectively.

Hence, rather than encouraging employees with lower skill levels to assess their own skills by reducing the fear of being blamed for bad outcomes, the treatment even reduced assessment participation particularly in this group. Assessment scores are somewhat reduced for the high-skill employees – likely because they can no longer impress their managers through good

assessment scores. To study the strength of such reputational concerns in more detail, we turn to the post-experimental survey where we included two items to assess different motives for taking part in the skill assessment (*"I participate to demonstrate to others that I am competent."*) or avoiding it (*"If I don't participate, it's because I expect to perform poorly."*). First of all, the motivation to demonstrate competence is substantially stronger than the fear of being viewed as incompetent. On a scale from 1 ("fully disagree") to 7 ("fully agree") the mean is 4.71 for the former and 2.19 for the latter. To see how the treatment affects these motives, we re-run the analyses above with the responses to each item as the dependent variables. As column (1) in Table 6 shows – and in line with the role of image concerns as a motivating force here –, participants in the treatment group are indeed significantly less motivated to participate in the knowledge assessment to demonstrate their ability to their manager. Column (2) shows that this effect is indeed driven by high-skill employees. In contrast to our prior expectations (but in line with our observations regarding the treatment effect on assessment participation), the treatment does not reduce the (already low) stated motive to avoid the skill assessment in fear of being exposed as incompetent (column 4 in Table 6). While low-skill employees indeed state significantly more often that they avoid the assessment due to being afraid to do poorly than high-skill employees (columns 3 and 4), there is no evidence for a differential treatment effect on this motive.

Table 6: Signaling and Reputational Concerns

	Demonstrate Competence		Fear of Incompetence	
	(1)	(2)	(3)	(4)
Treatment	-0.196** (0.078)	-0.004 (0.104)	0.066 (0.069)	0.102 (0.119)
Treatment × High-Skill		-0.392*** (0.144)		-0.040 (0.154)
High-Skill		0.131 (0.089)		-0.426*** (0.100)
Constant	0.090** (0.045)	0.030 (0.070)	-0.030 (0.042)	0.188** (0.072)
Wald-Test High-Skill		<0.001		0.491
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	125	125
Observations	850	785	847	782
R ²	0.010	0.021	0.001	0.050

Note: Table shows estimates from linear regressions on the strength of an employee's competence motif when considering participating in the skill assessment (columns 1-2) and the strength of an employee's avoidance motif when considering not participating in the skill assessment (columns 3-4). All dependent variables have been standardized. Questions are answered on a 7-point Likert scale from 1 = 'strongly disagree' to 7 = 'strongly agree'. Columns 2 and 4 report interaction models using the variable 'High-Skill', which takes value 1 for employees with an above-median average assessment score prior to the experiment and 0 otherwise. 'Wald-Test High-Skill' reports p-values from a Wald test of the null hypothesis that the treatment effect for high-skill employees is equal to zero (Treatment + Treatment × High-Skill = 0). Standard errors are clustered at the team level and shown in parentheses. ***, ** indicate significance at the 1% and 5% levels, respectively.

6.2 Manager Investment

In this section, we study the role of managers in the skill management process, which consists of scheduling development meetings and training. To assess the treatment effect on managerial attention, we first investigate how the treatment affected the likelihood that managers con-

ducted development meetings and at which frequency. We begin by regressing the meeting incidence (as reported by employees in the post-experimental survey) on a treatment indicator. As the results reported in Table 7 show, the treatment reduced the likelihood that at least one meeting is conducted from 0.77 to 0.67, i.e., by about 13.63% (see column 1). Hence, the intervention not only reduced employees' willingness to assess their own skill gaps but also managers' own time investments in employee development. To understand the behavioral drivers of this effect, it is informative to note that the treatment primarily affects the *likelihood* of conducting at least one meeting. As documented in column 3 of Table A10, the effect on the number of meetings has a similar magnitude which shows that it is the incidence rather than the frequency that differs. In other words, managers do not substantially reduce the meeting frequency *after* having held at least one meeting without having access to assessment outcomes. This suggests that managers are not reducing the frequency of meetings because they have had worse experiences as a result of the treatment. Instead, it appears that managers *ex ante* perceive the meetings as being less valuable when they know that they will no longer have access to hard information on employee skills. This further underlines the importance of hard information as a commitment device, not only for employees, but also for managers.

Next, we study how the reduction in meeting incidence affected employees with different prior skill levels. In the regressions reported in column 2 of Table 7, we investigate the differential effect of the treatment on meeting incidence for high- and low-skill employees. Interestingly, we find that the likelihood of having at least one development meeting is, in particular, substantially reduced among high-skill employees by 18.48% ($p = 0.011$). For low-skill employees, the negative treatment effect is less pronounced at 6.87% and not statistically significant.¹⁶ Apparently, as high-skill employees can no longer demonstrate their abilities, (as shown in Table 6) the managers also find it less rewarding to meet with them, for instance as it is harder

¹⁶This further shows that individual participation in the skill assessment does not appear to influence the likelihood of arranging development meetings, as the reduction in assessment participation is driven by *low-skill* employees. If assessment participation and managerial attention were directly correlated, the results shown in Table 7 would show the opposite effect. This is in line with our interpretation that managers make decisions on the allocation of managerial attention directly in response to the treatment.

to provide tangible recognition and have a fruitful discussion on their skill set.¹⁷ Similarly, as column 4 in Table 7 shows, the treatment reduced the training intensity in both groups, however the reduction appears to be larger for the high-skill ($p = 0.020$)

Given that high-skill employees appear to have lost a major source of recognition and training, it is conceivable that the overall reduction in job satisfaction is driven by these employees. Exploring this potential link, we indeed find that the negative effect on job satisfaction is larger in magnitude for high-skill employees (see Table A7 in the appendix). To understand whether this link stems from a lack of recognition or a lack of training, we conduct additional analyses to explore to what extent meeting incidence or the number of trainings mediate the treatment effect on job satisfaction (see Table A14 and Table A15). We caution that mediation analyses come with very strong assumptions, which are rarely satisfied, even in an experimental setting.¹⁸ As Table A14 shows, having at least one meeting during the treatment period is positively associated with job satisfaction and its inclusion in the regression reduces the treatment coefficient, whereas the number of trainings is not associated with job satisfaction conditional on treatment. The subsequent decomposition of the treatment effect indicates that approximately 20% of the treatment effect is mediated through development meetings (see Table A15). From this viewpoint, the negative effect on job satisfaction may be partially attributed to reduced interaction with the manager, again underlining the potential *positive* effects of performance monitoring on employee well-being.

¹⁷Note that the patterns reported in Table 6 are not driven by the reduction in meetings for high performers. The results remain virtually unchanged when controlling for meeting incidence or frequency as shown in Table A11.

¹⁸The causal interpretation of a mediator hinges on the assumption of sequential ignorability (Imai, Keele, & Tingley, 2010). Sequential ignorability requires potential outcomes to be a) independent of treatment status and b) observed and treatment-induced mediators to be independent from unobserved mediators. While the first condition holds in a randomized experiment by design, the second condition only holds if and only if the mediator is exogenous to the treatment status, i.e. independently randomized. As argued by Rohrer, Hünermund, Arslan, & Elson (2022) this condition is rarely satisfied, even in an experimental setting, so the results are to be interpreted with caution.

Table 7: Treatment effects on meetings and training

	Prob. of meeting		Num. of trainings	
	(1)	(2)	(3)	(4)
Treatment	−0.106** (0.043)	−0.052 (0.052)	−0.194** (0.091)	−0.171* (0.092)
Treatment × High-Skill		−0.091 (0.062)		−0.101 (0.066)
High-Skill		0.016 (0.039)		0.115** (0.053)
Constant	0.774*** (0.025)	0.761*** (0.032)	0.374*** (0.077)	0.348*** (0.077)
Wald-Test High-Skill		0.011		0.020
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	125	125
Observations	846	781	2582	2234
R ²	0.014	0.016	0.011	0.016

Note: Table shows estimates from linear regressions of the likelihood of at least one development meeting (columns 1-2) and the number of trainings (columns 3-4) on a treatment indicator. The dependent variables are a binary indicator that takes value 1 if an employee reported having had at least one meeting with their manager (columns 1-2) and the number of trainings per employee according to the HR records (columns 3-4). Columns 1 and 3 report full-sample specifications. Columns 2 and 4 report interaction models using the variable 'High-Skill', which takes value 1 for employees with an above-median average assessment score prior to the experiment and value 0 otherwise. 'Wald-Test High-Skill' reports p-values from a Wald test of the null hypothesis that the treatment effect for high-skill employees is equal to zero (Treatment + Treatment × High-Skill = 0). Standard errors are clustered at the team level and shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% levels, respectively.

7 Training Purpose and Work Performance

In the previous sections, we presented evidence showing that the treatment reduced skill assessment participation among low-skill employees, but reduced managerial attention and training in particular for the high-skill. This naturally raises the question of whether the reduction in the latter group is concerning for overall performance, given that skill levels are already high at the outset in this group. The analysis proceeds in two steps. We first study which employees and which types of training are most affected. We then examine whether performance declines in the assignments for which each type of training matters most. To begin with, it is important to consider the different purposes of training. As laid out in Section 2, the firm distinguishes between training to preserve existing skills and training to acquire new skills. When the reduction in training for high-skill employees is concentrated on the former, it may not be very detrimental or could even increase efficiency by eliminating useless time expenditures. If, however, the treatment affects the latter form of training, it may undercut the firm's ability to adapt and prepare its employees for changing skill requirements.

When managers register an employee for a training, they have to indicate whether the purpose of the training is to maintain an existing skill (referred to as *skill preservation*) or to develop a new skill (*new skill acquisition*).¹⁹ We can make use of this distinction to investigate which types of training are particularly affected in which employee group.²⁰ Figure 3 shows the average number of trainings, split by training type (skill preservation, new skill acquisition) and employee skill level (high, low). Table A13 shows the corresponding regression results. While training intensity is lower across both groups and training types, the effect is particularly pronounced for training dedicated to the acquisition of new skills among high-skill employees ($p = 0.01$, panel C) and, (to a lesser extent) for skill preservation training for the low-skill

¹⁹Managers can also select the option 'not relevant to a specific skill'. However, this option is rarely selected, so we drop this alternative from the analysis.

²⁰In Table A12, we investigate whether the overall number of trainings dedicated to skill preservation and acquisition is differentially affected by the treatment. We do not find such an effect. If anything, the treatment effect is slightly larger for training dedicated to the acquisition of new skills.

($p = 0.066$, panel B).

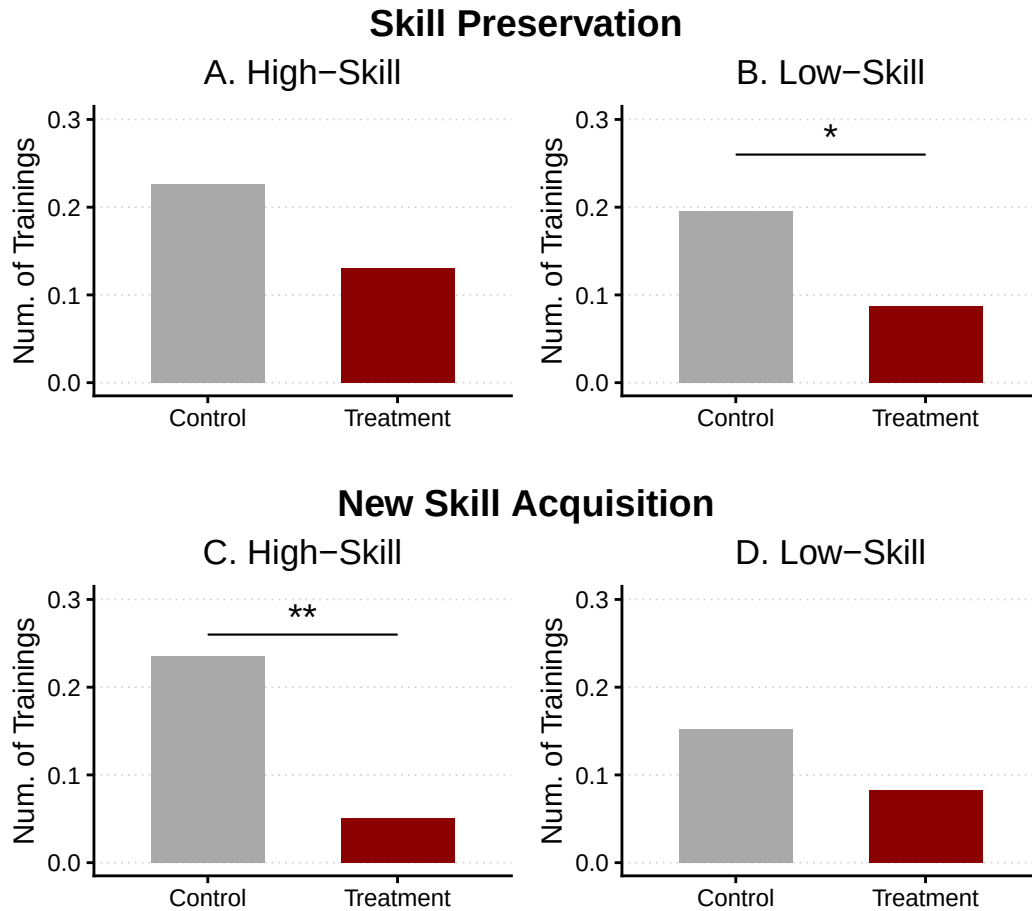


Figure 3: Treatment effects by Employee Type and Training Purpose

The observation that the reduction in trainings is particularly pronounced for new skill acquisition of high-skill employees may suggest that the overall reduction in work performance is concentrated on work assignments that require a higher level of skill or a broader skill set in general. To investigate this conjecture, we study heterogeneous treatment effects by the degree of assignment complexity. According to the firm's partition of work assignments into 13 different types, each type can be classified as either *routine* or *complex*. Routine assignments often include solving problems that occur on a regular basis or products that require little

technical sophistication to work with. Complex assignments often require non-standard custom solutions that have a higher risk of failure. These assignments are either associated with recent technologies that are inherently complex or a sophisticated problem that other technicians were previously unable to resolve. Therefore, assignments can be re-classified from routine to complex. In line with the firm's description, our final classification reveals that routine assignments are performed more regularly and take relatively little time ($M = 14.5$ min.), while complex assignments take more than twice as long ($M = 32.8$ min.), and occur less frequently (see Figure A8 for an anonymized classification of all work assignments). In Table 8, we run separate difference-in-differences regressions for tasks associated with either routine or complex work assignments with the average execution times as the dependent variable. As column (1) shows, there is no discernible treatment effect on the weekly average completion time for routine assignments. However, we indeed find a 3.169-minute increase in the weekly average completion time for complex tasks (column 2), which corresponds to a 9.65% increase relative to the pre-treatment average in the control group. In summary, we find that the reduction in training intensity is particularly strong for new skill acquisition among high-skill employees such that the reduction in active skill management in particular leads to a decline in up-skilling among these employees. In line with this pattern, the reduction in work performance is concentrated on complex work assignments.

Table 8: Treatment effects on Work Performance by Assignment Type

	Routine	Complex
Treatment	−0.044 (0.184)	3.169** (1.567)
Team FE	Yes	Yes
Task FE	Yes	Yes
Product × Week FE	Yes	Yes
Data Level	Team	Team
DV mean	14.53	32.84
Clusters	121	121
Observations	128 135	21 244
R ²	0.443	0.375

Note: Table shows average treatment effects from difference-in-differences regressions on the average time per task in minutes. Column (1) shows results for work assignments classified as routine assignments, whereas column (2) shows results for complex assignments. For all regressions, we focus on the four key tasks used in the previous analysis. All models include Team, Task, as well as Product × Week fixed effects. Pre-treatment averages from the control group are depicted below the estimates. Standard errors are clustered at the team level and shown in parentheses. ** indicates significance at the 5% level.

8 Conclusion

We studied the value of skill management in organizations through a natural field experiment in a large organization. Our findings highlight that active skill monitoring by supervisors is crucial for human capital investments and firm performance. Even a slight reduction in managers' ability to monitor employee skills significantly decreased training intensity and productivity. Additionally, contrary to existing literature in psychology and behavioral economics, we observed that reduced manager monitoring also led to a decline in job satisfaction. Our results therefore suggest that monitoring is not simply a surveillance tool, but may also improve both productivity and employee satisfaction at the same time.

We further studied the underlying behavioral mechanisms and the roles of employees and

managers in this process. Reducing managers' ability to monitor employee skills diminished employees' willingness to assess their own skill gaps, particularly among low-skill employees. Instead of encouraging these employees to identify training needs without fear of blame, the removal of monitoring had the opposite effect. This result highlights the significance of moral hazard issues in training investments. Despite the benefits of human capital gains for employees, there is thus a tendency to under-invest, and active skill management helps to mitigate this problem. Moreover, we find that the removal of access to information on employee skill levels also reduced the likelihood that managers initiate development meetings with their employees. Granting managers access to hard information about employee skills thus (i) commits employees to invest in their own learning and (ii) fosters managers' own time investments in employee development.

Having documented the main effects and general mechanics of skill management, we set out to understand the connection between the assignment of training and team productivity in more detail. In particular, we study the effects of the intervention for employees of different skill levels. Sandvik et al. (2026), for instance, found that uptake of voluntary training is lower among low-performers, even though they benefit from training the most. In environments undergoing technological changes, it is, however, less clear whether prioritizing the skill gaps of low-skill employees should take precedence over broadening the skill set of high-skill employees, as firms constantly need to adapt to changing skill requirements. Our analysis indeed shows that less extensive skill management induced by the treatment particularly reduced managerial attention and subsequent training intensity also among high-skill employees. Among high-skill employees, the largest estimated reduction concerns training intended to support the acquisition of new skills. In turn, the intervention led to a reduction in work performance in complex, non-routine tasks. In summary, our results demonstrate that active skill management and close managerial scrutiny of skill levels are crucial for organizational performance. They foster the willingness of low-skill employees to assess their own skills, encourages high-skill

employees to improve and acquire new skills, and ensures that leaders invest time in helping both groups strengthen their human capital.

We close by considering the extent to which our findings are likely to generalize beyond our specific setting. In the setting we study, employees perform highly skilled work in a technologically dynamic environment, which requires almost continuous investment in employees' human capital. In contrast to that, some industries require less (continuous) training investment and should not be as sensitive to changes in the skill management process. However, Diaz et al. (2025) show that engaging with employees and supporting their development increases performance across different industries, such as retail, manufacturing, and quick-service restaurants. Further research could identify the institutional characteristics that determine the effectiveness of a particular skill development strategy (e.g. broad vs. specific). In addition to that, turnover is rather low in our setting as the firm we study holds a strong market position, likely allowing the firm to capture large parts of the returns to training. In environments with higher turnover — often driven by more competitive labor markets — employers may be less able to capture the returns to the training they provide before an employee leaves. Ahimbisibwe, Brownback, Hirshleifer, & Rezaee (2026), for instance, show that this ability to extract returns from training is a key determinant of employer demand for training. Future studies could therefore examine the interplay between organizational investments in training and local labor market conditions. Finally, our field service technicians work largely alone and interact little with managers day-to-day. In more collaborative or co-located settings (e.g., regular office jobs), managers may already learn about employees' skills through ordinary interaction, reducing the need for explicit monitoring via a learning platform. How management practices support employee development across such work environments — remote versus in-person — thus remains an open question for future research.

Taken together, these boundary conditions point to where our findings are likely to matter most: in settings where technical requirements evolve quickly and human capital drives

firm performance, managerial access to hard information on employee skills is not merely a monitoring tool — it is what aligns employee and managerial effort toward continuous learning.

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A Appendix

A.1 Conceptual Framework

First consider a simple conceptual framework which models the tension between managerial support and control in skill management. A supervisor S and agent A can both invest time to prepare an investment in the agent's human capital. The agent can take an action $p \in \{0, 1\}$ to learn about own skill deficiencies by participating in a skill assessment. Simultaneously the supervisor S can invest in supporting the agent's human capital development by setting up a development meeting $m \in \{0, 1\}$ to talk about training needs. Taking an action causes an idiosyncratic private cost c_i for each party $i \in \{A, S\}$ which are drawn from a uniform distribution $U[0, C_i]$ and we assume that C_S and C_A are sufficiently large that there are always types who find it too costly to invest. The agent's prior skill level $\theta \in \{H, L\}$ is known to both parties. The key treatment intervention is to provide supervisors with a skill monitoring technology providing them with access to the skill assessment outcomes, denoted by the binary treatment indicator $a \in \{0, 1\}$.

Let the expected resulting human capital increase be $h_a(p, m, \theta)$, which is higher when the agent has invested in the skill assessment ($p = 1$) and the meeting is scheduled ($m = 1$) and both are complements (i.e. each party's investment is more valuable when the other party invests as well) such that:

$$\Delta_a(\theta) = h_a(1, 1, \theta) - h_a(1, 0, \theta) - h_a(0, 1, \theta) + h_a(0, 0, \theta) > 0$$

Moreover, supervisor access to the assessment outcomes can only matter when the agent participated and the supervisor scheduled a meeting (i.e. $h_1(1, 1, \theta) > h_0(1, 1, \theta)$ and $h_1(p, m, \theta) = h_0(p, m, \theta)$ for $p + m < 2$).

Moreover, when a meeting takes place in which supervisors have access to the skill assessment outcome (i.e. $m = p = a = 1$) then the agent also has a reputational payoff v_θ with

$v_H > 0 > v_L$. That is, the interaction will be pleasant in expectation if the prior skill level is high as then will consists of giving/receiving praise, but otherwise will be unpleasant due to the fear or embarrassment of discussing negative test outcomes. The supervisor's utility is:

$$h_a(p, m, \theta) - c_S$$

and the agent's is:

$$h_a(p, m, \theta) + a\phi_A v_\theta - c_A$$

where ϕ_A measures the weight the agent attaches to the reputational payoff.

Let now $\delta_p(\theta) = h_a(1, 0, \theta) - h_a(0, 0, \theta) > 0$ and $\delta_m(\theta) = h_a(0, 1, \theta) - h_a(0, 0, \theta) > 0$ be the baseline human capital gains from each party's action (without the respective other's investment) and note that these baseline gains are independent from a .

A.2 Equilibrium Analysis

Let $\rho_p = \Pr(p = 1)$ and $\rho_m = \Pr(m = 1)$ and given the distribution of cost parameters we have that:

$$\rho_m(\rho_p) = \max \left\{ \frac{\delta_m(\theta) + \rho_p \Delta_a(\theta)}{C_S}, 0 \right\} \quad (1)$$

and

$$\rho_p(\rho_m) = \max \left\{ \frac{\delta_p(\theta) + \rho_m(\Delta_a(\theta) + \phi_A a v_\theta)}{C_A}, 0 \right\} \quad (2)$$

When there is an internal solution with $\rho_m, \rho_p > 0$ this leads to:

$$\rho_m(a, \theta) = \frac{C_A \delta_m(\theta) + \delta_p(\theta) \Delta_a(\theta)}{C_A C_S - (\Delta_a(\theta) + \phi_A a v_\theta) \Delta_a(\theta)}$$

$$\rho_p(a, \theta) = \frac{C_S \delta_p(\theta) + \delta_m(\theta) (\Delta_a(\theta) + \phi_A a v_\theta)}{C_A C_S - (\Delta_a(\theta) + \phi_A a v_\theta) \Delta_a(\theta)}$$

We can now analyze how the equilibrium depends on the granted access a depending on the agent's type. For a high-skill agent ($\theta = \theta_H$), discussing the assessment yields praise, such that skill monitoring increases both human capital gains and social and reputational returns. Under these conditions, we are in a regime of strategic complementarities, where joint action is mutually reinforcing – such that there is no tension between the supportive and the controlling aspect of skill monitoring. We have that $\rho_m(1, \theta) > \rho_m(0, \theta)$ and $\rho_p(1, \theta) > \rho_p(0, \theta)$ and thus:

Proposition A.1. *For high-skill agents test participation and meeting frequency are always higher under skill monitoring (when $a = 1$).*

However, for a low-skill agent ($\theta = \theta_L$), the expectation of an embarrassing discussion introduces a potentially negative reputational payoff. If the agent's reputational concerns are sufficiently strong, this social penalty can overwhelm the human capital gains as the following result shows:

Proposition A.2. *For low-skill agents test participation and meeting frequency are strictly higher under skill monitoring if and only if ϕ_A is not too large.*

Proof. Let $A_a(\phi_A) = \Delta_a(L) + a\phi_A v_L$. For interior solutions, the equilibrium rates are given by:

$$\rho_p^L = \frac{C_S \delta_p(L) + \delta_m(L) A_a(\phi_A)}{C_A C_S - A_a(\phi_A) \Delta_a(\theta)}$$

$$\rho_m^L = \frac{C_A \delta_m(\theta) + \delta_p(L) \Delta_a(\theta)}{C_A C_S - A_a(\phi_A) \Delta_a(\theta)}$$

At $\phi_A = 0$, we have that $A_1(0) = \Delta_1(L) > \Delta_0(L) = A_0(0)$. Thus, without fear of embarrassment, monitoring strictly increases both test participation ρ_p^L and through strategic

complementarity also meeting frequency ρ_m^L .

Under no-monitoring ($a = 0$), we have that $A_0(\phi_A) = \Delta_0(L) > 0$ for $\forall \phi_A$ and thus:

$$\rho_p^L = \frac{C_S \delta_p(L) + \delta_m(L) \Delta_0(L)}{C_A C_S - (\Delta_a(L))^2} > 0$$

$$\rho_m^L = \frac{C_A \delta_m(L) + \delta_p(L) \Delta_0(L)}{C_A C_S - (\Delta_a(L))^2} > 0$$

Under monitoring ($a = 1$) $A_1(\phi_A)$ is strictly decreasing in ϕ_A and so are ρ_p^L and ρ_m^L . As $A_1(\phi_A) \rightarrow -\infty$ we have that $\rho_p^L(a, \phi_A) = 0$ for ϕ_A sufficiently large and, in turn:

$$\rho_m = \frac{\delta_m(\theta)}{C_S}$$

The result then follows from the intermediate value theorem. $\square$

When reputational concerns become sufficiently strong for the agents, test participation is reduced under monitoring. Here the agent's fear of embarrassment can outweigh human capital gains when the manager has access to skill assessment outcomes. In turn, the meeting becomes less valuable and the managers also conduct these meetings less often. As a result, removing monitoring may encourage those agents to assess their skills and can lead to higher human capital gains.

A.3 Tables

Given that the number of trainings an employee receives in a given time frame can be modeled as count data, we consider alternative specifications for the main analyses conducted in Table 2. For the analysis conducted in Table A1, we consider a Pseudo-Poisson-ML estimator, which performs well on over-dispersed count data (Santos Silva & Tenreyro, 2006). Estimations are performed in R using the *fixest* package (Bergé, 2018). To interpret the results from columns (1) and (2) of Table A1, we calculate the incidence-rate ratio by exponentiating the treatment coefficients. Overall, we find that the results are similar to those reported in the main text.

For the baseline treatment comparison without covariates, the incidence-rate ratio is 0.481, which is a 51.89% reduction in the number of trainings. Including prior participation in and scores of the skill assessment changes the incidence-rate ratio to 0.479, which is equal to a 52.08% reduction in the number of trainings. In the next step, we re-analyze the treatment effect on whether or not an employee receives any training at all. We find that employees in the treatment group have an odds ratio of 0.49 corresponding to 51.04% lower odds of receiving at least one training during the treatment period. Finally, columns (5) and (6) show team-level results, without (column 5) and including pre-treatment data (column 6). While we focus our analyses on the key skills almost all employees possess and for which we have pre- and post-treatment data, we further analyzed treatment effects on the remaining skill assessments that took place in the treatment period. As Table [A4](#) shows, the reduction in assessment participation is strongest among the most widespread skills. In this table, the number of observations denotes all employees that were assigned to the knowledge assessment at the time it took place. As outlined previously, the number of observations is much smaller for less widespread skills. However, including all skills into the analysis does not change our results.

Table A1: Treatment Effects on Training (Robustness Checks)

	Num. of Trainings		Prob. of Training		Num. of Trainings	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	−0.732** (0.340)	−0.736** (0.355)	−0.714** (0.328)	−0.690** (0.336)	−0.693** (0.329)	−0.682** (0.329)
Prior Participation		1.196** (0.533)		1.413*** (0.505)		
Prior Training						0.713* (0.407)
Prior Score		0.066 (0.059)		0.103 (0.075)		
Constant	−0.983*** (0.207)	−1.963*** (0.526)	−1.702*** (0.205)	−2.873*** (0.516)	−0.828*** (0.201)	−1.034*** (0.250)
Data Level	Individual	Individual	Individual	Individual	Team	Team
Model	Poisson	Poisson	Logit	Logit	Poisson	Poisson
Observations	2582	2078	2582	2078	125	125
Clusters	125	125	125	125	125	125

Note: Table shows estimates from pseudo-poisson regressions on the number of trainings as a dependent variable, which is the sum of all trainings associated with key skills (columns (1), (2), (5) and (6)). In columns (3) and (4) the dependent variable is a binary indicator for receiving training or not that takes value 1 if an employee has received at least one training and value 0 otherwise. In columns (5) and (6), we use the average number of trainings per employee across the team as the dependent variable. Prior participation captures the participation rate prior to the experiment. Prior score denotes the average standardized ratio of correctly answered questions on a skill assessment. Prior training denotes the number of trainings per team member prior to the experiment. Robust standard errors (clustered at the team level in columns 1-4) are shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A2: Treatment Effects on Quality and Customer Satisfaction

	Unresolved Visit Rate		Customer Satisfaction	
	(1)	(2)	(3)	(4)
Treatment	0.002 (0.002)	0.002 (0.002)	0.016 (0.014)	0.016 (0.014)
Team FE	Yes	Yes	Yes	Yes
Week FE	Yes	Yes	Yes	Yes
Region FE	No	Yes	No	Yes
Data Level	Team	Team	Team	Team
DV mean	0.10	0.10	4.67	4.67
Clusters	121	121	121	121
Observations	9170	9170	7101	7101
R ²	0.186	0.186	0.053	0.053

Note. Table shows average treatment effects from difference-in-differences regressions on the unresolved visit rate and average customer satisfaction. The unresolved visit rate denotes the proportion of customer service orders that are not resolved during the initial technician visit and require additional follow-up actions or appointments. Customer Satisfaction is measured with a short online survey sent to the customer after a visit. Robust standard errors are shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A3: Employee and Manager Investments in the Development Process

	Number of Trainings	
	(1)	(2)
Treatment	−0.104 (0.140)	−0.102 (0.139)
Test Participation	0.349*** (0.118)	0.042 (0.110)
Meeting Conducted	0.229** (0.111)	−0.121 (0.140)
Meeting Conducted × Test Participation		0.427** (0.190)
Constant	−0.035 (0.150)	0.217 (0.137)
Data Level	Individual	Individual
Clusters	125	125
Observations	797	797
R ²	0.022	0.025

Note. Table shows estimates from linear regressions on the number of trainings per employee. In column (1), we regress the number of trainings on a treatment dummy, the share of tests an employee has completed during the treatment period, and a binary indicator for employee development meetings that takes value 1 if an employee indicated in the post-experimental survey that they had at least one development meeting in the last 6 months and 0 otherwise. In column (2), we include an interaction term between the meeting indicator variable and the share of tests an employee has completed during the treatment period. Standard errors are clustered at the team level and shown in parentheses. ***, ** indicate significance at the 1% and 5% levels, respectively.

Table A4: Treatment Effects on Assessment Participation (All Skills)

	Skill 1	Skill 2	Skill 3	Skill 4	Skill 5	Skill 6	All Skills
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treatment	−0.094*** (0.023)	−0.046* (0.025)	0.001 (0.046)	−0.035 (0.053)	−0.034 (0.041)	−0.147** (0.069)	−0.066*** (0.022)
Constant	0.855*** (0.014)	0.793*** (0.018)	0.649*** (0.037)	0.646*** (0.039)	0.897*** (0.029)	0.931*** (0.031)	0.783*** (0.015)
Data Level	Individual	Individual	Individual	Individual	Individual	Individual	Individual
Clusters	125	125	125	121	107	71	125
Observations	2286	2032	1068	450	341	109	2329
R ²	0.014	0.003	0.000	0.001	0.003	0.045	0.009

Note: Table shows estimates from linear regressions of participation in each skill assessment during the treatment period on a treatment indicator (columns 1-6) and the average of all post-treatment assessments (column 7). Standard errors are clustered at the team level and shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A5: Treatment effects on assessment participation and scores

	Assessment Participation		Assessment Scores	
	(1)	(2)	(3)	(4)
Treatment	−0.042** (0.020)	−0.041** (0.020)	−0.022 (0.051)	−0.016 (0.044)
Prior Participation	0.376*** (0.032)	0.375*** (0.032)	0.377*** (0.074)	0.341*** (0.073)
Prior Part. × Prior Score		0.034*** (0.008)		0.597*** (0.025)
Constant	0.473*** (0.032)	0.475*** (0.032)	−0.354*** (0.066)	−0.313*** (0.066)
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	124	124
Observations	2232	2232	1988	1988
R ²	0.111	0.117	0.012	0.283

Note: Table shows estimates from linear regressions on assessment participation (columns 1 and 2) and assessment scores (columns 3 and 4). Participation denotes the number of assessments an employee has taken part in during the treatment period, divided by the number of assignments they have been assigned to. Assessment scores depict the average standardized ratio of correctly answered questions on the knowledge assessment. Prior participation captures the participation rate prior to the experiment. Prior participation × prior score is a multiplicative term of the participation rate prior to the experiment and the average standardized ratio of correctly answered questions in all skill assessments prior to the experiment. Standard errors are clustered at the team level and shown in parentheses. **, *** indicate significance at the 5% and 1% levels, respectively.

Table A6: Pre-treatment characteristics of survey participants

	Control (N=503)		Treatment (N=420)		Diff. in Means	p
	Mean	Std. Dev.	Mean	Std. Dev.		
Assessment Participation	0.88	0.24	0.87	0.27	−0.01	0.69
Assessment Score	0.05	0.84	0.01	0.86	−0.04	0.48

Note: Table shows pre-experimental data for employees that participated in the post-experiment survey, split by treatment group. Assessment participation denotes the average participation for the tests every member of the team has been assigned to. Assessment results depict the average standardized ratio of correctly answered questions on a knowledge test. P-values are calculated using OLS, regressing the dependent variable on a treatment indicator.

Table A7: Treatment Effects on Job Satisfaction by Employee Skill Level

	Job Satisfaction	
	(1)	(2)
Treatment	−0.200** (0.077)	−0.115 (0.118)
Treatment × High-Skill		−0.174 (0.156)
High-Skill		0.224* (0.114)
Constant	0.094* (0.053)	−0.025 (0.085)
Data Level	Individual	Individual
Wald-Test High-Skill		0.01
Clusters	125	125
Observations	775	713
R ²	0.010	0.018

Note: Table shows estimates from linear regressions on job satisfaction as reported by the employees in a post-experimental survey. Job satisfaction is standardized and measured using a single-item measure from the Linked Personnel Panel (Kampkoetter et al., 2016) and answered on a 7-point Likert scale from 1 = 'Strongly disagree' to 7 = 'Strongly agree'. Column 2 reports the interaction model using the variable 'High-Skill', which takes value 1 for employees with an above-median average assessment score prior to the experiment and value 0 otherwise. 'Wald-Test High-Skill' reports p-values from a Wald test of the null hypothesis that the treatment effect for high-skill employees is equal to zero (Treatment + Treatment × High-Skill = 0). Standard errors are clustered at the team level and reported in parentheses. *, ** indicate significance at the 10% and 5% levels, respectively.

Table A8: Treatment effects on task completion times (Quarterly Results)

	Customer Service	Troubleshooting	Quality Control	Documentation
	(1)	(2)	(3)	(4)
Treatment	1.886** (0.855)	3.755** (1.663)	−0.145 (0.984)	0.067 (0.372)
Team FE	Yes	Yes	Yes	Yes
Type × Quarter FE	Yes	Yes	Yes	Yes
Data Level	Team	Team	Team	Team
DV mean	6.27	43.29	10.22	4.48
Clusters	121	121	121	121
Observations	4150	4676	4635	4455
R ²	0.138	0.504	0.297	0.331

Note: Table shows average treatment effects from difference-in-differences regressions on the average time per task in minutes. All models include team as well as Type × Quarter fixed effects. Assignment types are usually associated with a specific technology, but do vary with respect to their degree of complexity. Pre-experimental averages from the control group are depicted in the 'DV mean' column of the table. We exclude four teams from the analyses for which no post-treatment performance data was available. Standard errors are clustered at the team level and reported in parentheses. *, **, indicate significance at the 10% and 5% levels, respectively.

Table A9: Reasons for Skill Assessment Participation

	Obligation	Demand
	(1)	(2)
Treatment	−0.011 (0.063)	−0.026 (0.069)
Constant	0.005 (0.044)	0.012 (0.046)
Data Level	Individual	Individual
Clusters	125	125
Observations	850	850
R ²	0.000	0.000

Note: Table shows results from linear regressions on standardized responses from the post-experimental survey. Columns capture the perceived obligation to take the test ('I participate in the assessment because it is part of my job.') and manager demand ('I participate in the assessment because my supervisor expects me to do it.'). Standard errors are clustered at the team level and reported in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A10: Treatment effects on meetings (Robustness Checks)

	Prob. of meeting		Num. of meetings	
	(1)	(2)	(3)	(4)
Treatment	−0.106** (0.043)	−0.131*** (0.044)	−0.148* (0.084)	−0.160* (0.090)
Prior Meetings		0.231*** (0.058)		0.444*** (0.119)
Constant	0.774*** (0.025)	0.539*** (0.061)	1.189*** (0.054)	0.812*** (0.116)
Data Level	Individual	Team	Individual	Team
Clusters	125		125	
Observations	846	95	846	95
R ²	0.014	0.232	0.006	0.169

Note: Table shows estimates from linear regressions of the likelihood or frequency of development meetings on a treatment indicator. For the individual-level analyses, the dependent variables are a binary indicator that takes value 1 if an employee reported having had at least one meeting with their manager (1) and the overall number of meetings (3), which can take values between 0 and 3. For the team-level analysis, the dependent variables are the share of employees with at least one meeting per team (2) and the average number of meetings summarized across each team (4). Columns (2) and (4) further include the number of meetings per team as a control variable, which was assessed in the pre-experimental survey. Not every team met the minimum participation threshold of 5 responses, which is why several teams could not be included in this analysis. Re-running the team-level analyses without control variables does not change the results. Robust standard errors (clustered at the team level in columns 1 and 3) are shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% level, respectively.

Table A11: Signaling and reputational concerns (controlling for meeting incidence)

	Demonstrate Competence		Fear of Incompetence	
	(1)	(2)	(3)	(4)
Treatment	-0.292*	0.053	0.086	0.158
	(0.150)	(0.208)	(0.107)	(0.184)
Treatment $\times$ High-Skill		-0.723**		-0.082
		(0.292)		(0.238)
High-Skill		0.262		-0.658***
		(0.186)		(0.154)
Meeting Conducted	0.985***	0.956***	-0.143	-0.094
	(0.158)	(0.168)	(0.119)	(0.121)
Constant	4.131***	4.038***	2.240***	2.538***
	(0.150)	(0.181)	(0.111)	(0.139)
Wald-Test High-Skill		0.002		0.585
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	125	125
Observations	846	781	843	778
R ²	0.055	0.065	0.003	0.052

Note: Columns (1) and (2) show estimates from linear regressions on the strength of an employee's competence motif when considering taking part in the skill assessment. Columns (3) and (4) show estimates from linear regressions on the strength of an employee's avoidance motive when considering not to participate in the skill assessment. All questions are answered on a 7-point Likert scale from 1 = 'Strongly disagree' to 7 = 'Strongly agree'. The variable 'High-Skill' takes value 1 for all employees with an above median average assessment score prior to the experiment and value 0 otherwise. 'Wald Test High-Skill' Wald test of the null hypothesis that the treatment effect for high-skill employees is equal to zero (Treatment + Treatment $\times$ High-Skill = 0). Standard errors are clustered at the team level and shown in parentheses. **, *** indicate significance at the 5% and 1% levels, respectively.

Table A12: Treatment Effects on Training by Type

	Num. of Trainings		Prob. of Training	
	(1)	(2)	(3)	(4)
Treatment	−0.101** (0.045)	−0.108** (0.051)	−0.052** (0.023)	−0.055** (0.024)
Treatment × Skill Preservation		0.015 (0.065)		0.005 (0.027)
Skill Preservation	0.032 (0.032)	0.024 (0.054)	0.023* (0.013)	0.020 (0.021)
Constant	0.171*** (0.040)	0.175*** (0.044)	0.087*** (0.019)	0.088*** (0.020)
Clusters	125	125	125	125
Observations	5164	5164	5164	5164
Wald-Test Skill Preservation		0.12		0.09
Data Level	Individual	Individual	Individual	Individual
R ²	0.008	0.008	0.012	0.012

Note: Table shows estimates from linear regressions on employee training. In columns (1) and (2) the dependent variable is the number of trainings, while columns (3) and (4) cover the probability of training. Columns (1) and (3) report main effects, while columns (2) and (4) include an interaction between treatment and skill preservation. Skill preservation is a binary indicator equal to 1 if the training purpose is the preservation of an existing skill and 0 otherwise. “Wald-Test Skill Preservation” reports the p-value from a Wald test of the null hypothesis that the treatment effect for skill-preservation training equals zero. Robust standard errors clustered at the team level are shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% level, respectively.

Table A13: Treatment effects by Training Purpose and Employee Type

	Skill Preservation		New Skill Acquisition	
	High-Skill	Low-Skill	High-Skill	Low-Skill
Treatment	−0.095 (0.076)	−0.108* (0.058)	−0.185** (0.071)	−0.070 (0.050)
Constant	0.226*** (0.057)	0.196*** (0.051)	0.235*** (0.066)	0.153*** (0.040)
Data Level	Individual	Individual	Individual	Individual
Clusters	125	125	125	125
Observations	1043	1191	1043	1191
R ²	0.006	0.010	0.018	0.004

Note: Table shows estimates from linear regressions on training devoted to skill preservation (columns 1 and 2) and new skill acquisition (columns 3 and 4). The variable 'High-Skill' takes value 1 for all employees with an above median average assessment score prior to the experiment and value 0 otherwise. Standard errors are clustered at the team level and shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% level, respectively.

Table A14: Mediation (Causal Steps)

	Job Satisfaction		
	(1)	(2)	(3)
Treatment	-0.200** (0.077)	-0.161** (0.076)	-0.205*** (0.076)
Meeting Conducted		0.385*** (0.086)	
Number of Trainings			-0.032 (0.039)
Constant	0.094* (0.053)	-0.206** (0.090)	0.107** (0.053)
Clusters	125	125	125
Observations	775	775	775
Data Level	Individual	Individual	Individual
R ²	0.010	0.039	0.011

Note: Table shows estimates from linear regressions on job satisfaction (standardized), measured using a single-item measure from the Linked Personnel Panel (Kampkoetter et al., 2016). *, **, *** indicate significance at the 10%, 5% and 1% level, respectively.

Table A15: Decomposition of the Treatment Effect

Effect	Estimate	Share of Effect
Total treatment effect on job satisfaction	-0.200***(0.077)	
Direct treatment effect on job satisfaction	-0.161** (0.075)	80.1
Effect mediated by development meetings	-0.040** (0.019)	19.9

Note. Decomposition of the treatment effect on job satisfaction into the direct effect and the indirect effect mediated by reporting at least one development meeting with the manager in the last 6 months. Values in parentheses indicate standard errors clustered at the team level. This analysis includes the indirect path through meetings, but not the indirect path through the number of trainings. *, **, *** indicate significance at the 10%, 5% and 1% levels, respectively.

Table A16: Main Outcomes (Excluding Teams with SUTVA violations)

	Num. of Trainings	Test Participation	Prob. of Meeting	Job Satisfaction	Customer Service	Troubleshoot- ing
Treatment	−0.212** (0.096)	−0.069*** (0.023)	−0.105** (0.044)	−0.177** (0.078)	0.503* (0.261)	1.683** (0.762)
Constant	0.392*** (0.083)	0.793*** (0.015)	0.774*** (0.026)	0.085 (0.055)		
Clusters	121	121	121	121	117	117
Observations	2490	2245	815	749	32 678	41 375
R ²	0.013	0.010	0.014	0.008	0.087	0.267
Data Level	Individual	Individual	Individual	Individual	Team	Team

Note: Table shows estimates from linear regressions on the number of trainings (column 1), average test participation (column 2), the probability of having at least one development meeting (column 3), job satisfaction measured using a one-item variable from Kampkoetter et al. (2016) (z-scored, column 4), the weekly average number of minutes devoted to customer service (column 5), and troubleshooting (column 6), respectively. For all estimations, teams where at least one member has joined from a different treatment arm are excluded from the analysis, which leads to the removal of 4 teams (3 teams from the control group, 1 team from the treatment group). Standard errors are clustered at the team level and shown in parentheses. *, **, *** indicate significance at the 10%, 5% and 1% level, respectively.

Table A17: Treatment effects on employee perceptions (with Lee Bounds)

	Job Satisfaction	Supervisor Support	Turnover Intention
	(1)	(2)	(3)
Treatment	−0.200** (0.077)	−0.214** (0.097)	0.088 (0.079)
Constant	0.094* (0.053)	0.098 (0.062)	−0.040 (0.055)
Data Level	Individual	Individual	Individual
Lee Bounds	[−0.513, −0.200]	[−0.491, −0.178]	[0.088, 0.433]
Clusters	125	125	125
Observations	775	848	819
R ²	0.010	0.011	0.002

Note: Table shows estimates from linear regressions on standardized measures of job satisfaction, supervisor support and turnover intentions as reported by the employees in the post-experimental survey. Job satisfaction and turnover intentions are measured using single-item measures from the Linked Personnel Panel (Kampkoetter et al., 2016), our measure of supervisor support is based on a four-item survey from Decius, Schaper, and Seifert (2021) and a three-item scale from Rimann and Udris (1999). All questions are answered on a 7-point Likert scale from 1 = ‘strongly disagree’ to 7 = ‘strongly agree’. Standard errors are clustered at the team level and reported in parentheses. Generalized Lee bounds (Semenova, 2025) are reported in square brackets. *, ** indicate significance at the 10% and 5% levels, respectively.

A.4 Figures

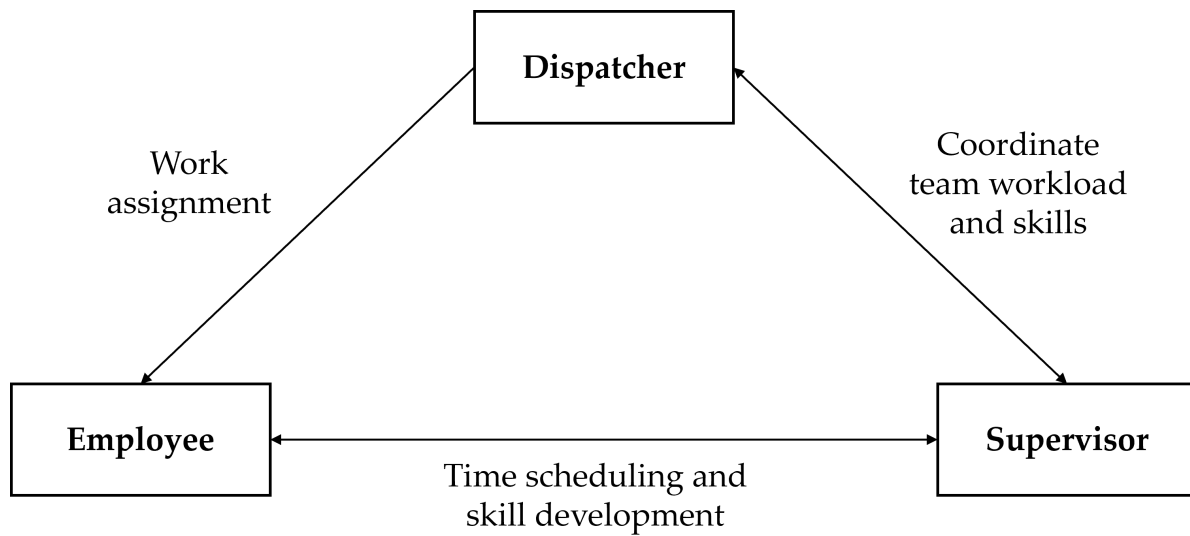


Figure A1: Responsibilities in the Firm

Control	Treatment
What would help you to get even better? Describe in detail which measures would help you to further develop your skills. You can also use our suggestions below. Then submit your ideas. Once you have sent them, they will be sent directly to your supervisor. Talk to your supervisor about your suggestions in the development dialog. Your Ideas What would help you to get even better?	What would help you to get even better? Describe in detail which measures would help you to further develop your skills. You can also use our suggestions below. Then submit your ideas. Once you have sent them, they will be sent directly to your supervisor. Talk to your supervisor about your suggestions in the development dialog. IMPORTANT: Do not bring your results sheet to the meeting. Your supervisor will focus on your ideas for further development and not ask for your test results. Your Ideas What would help you to get even better?

Figure A2: Communicating the treatment on the skill assessment platform

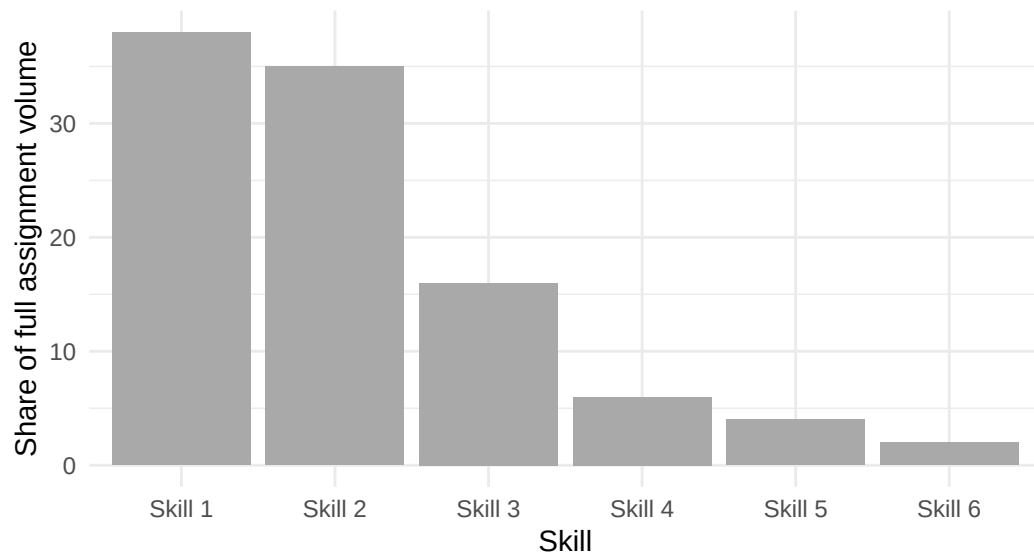


Figure A3: Relative share of work assignment volume by skill

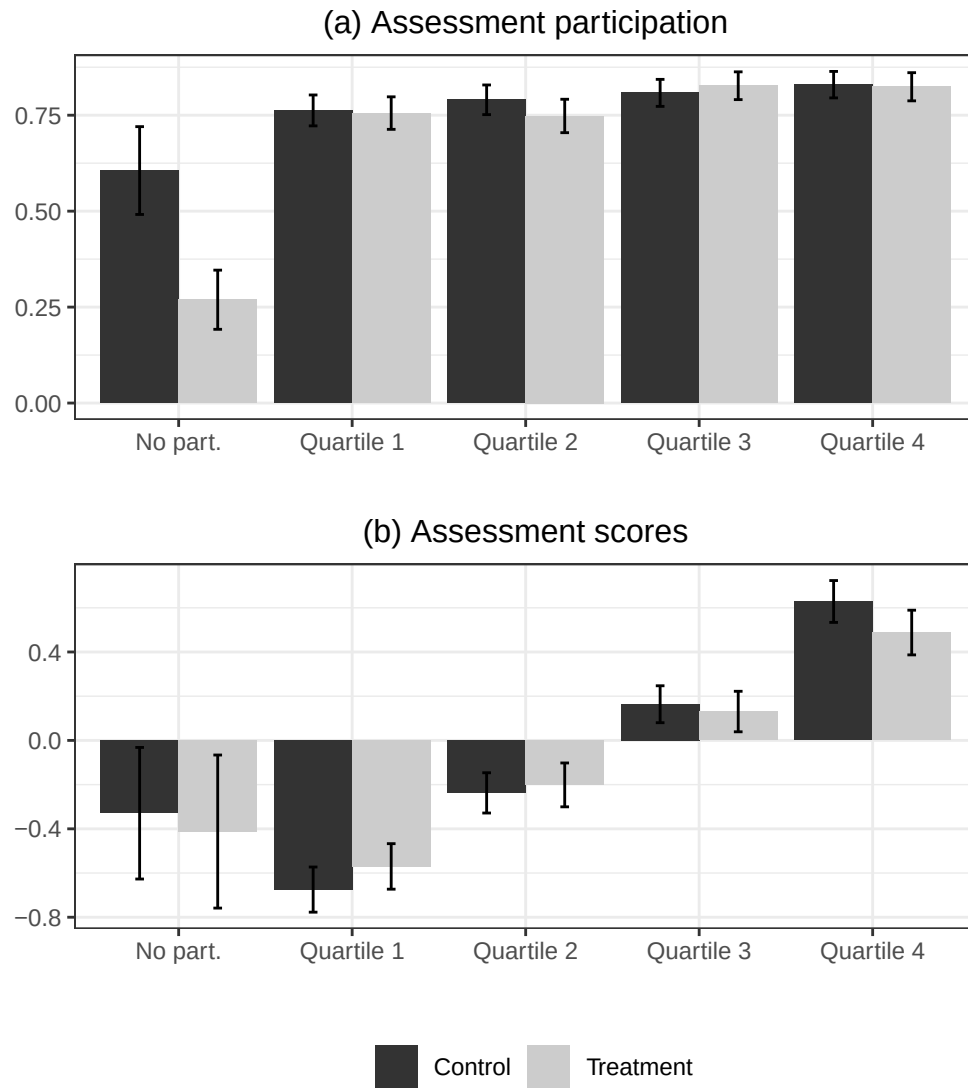


Figure A4: Outcomes by prior test participation and performance

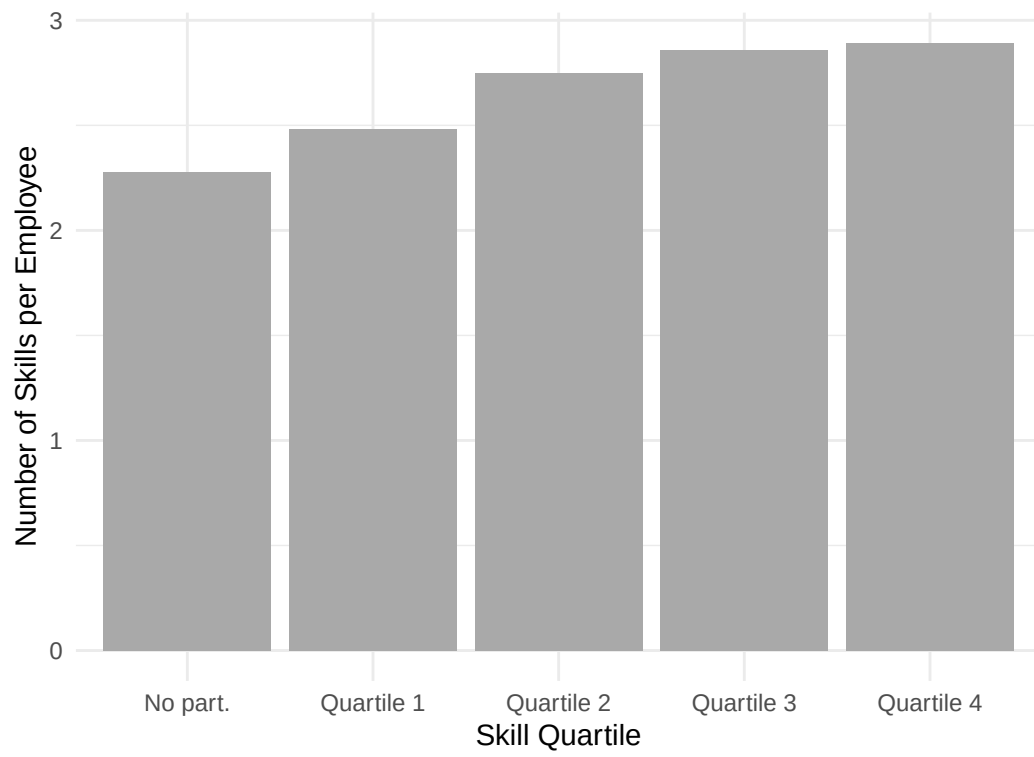


Figure A5: Average number of skills per quartile

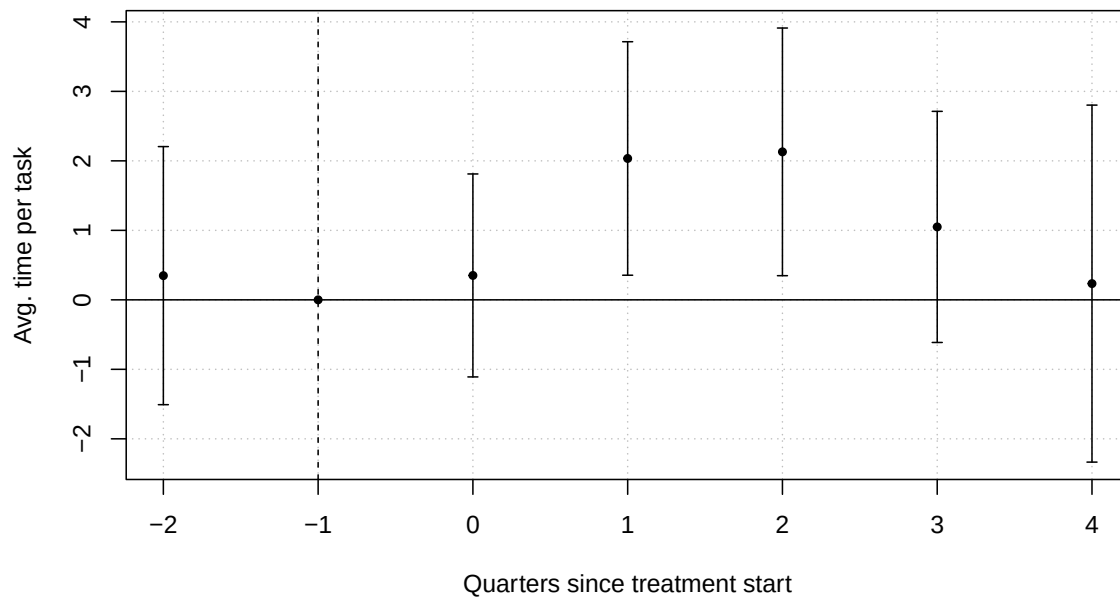


Figure A6: Treatment Effect on Customer Service Times

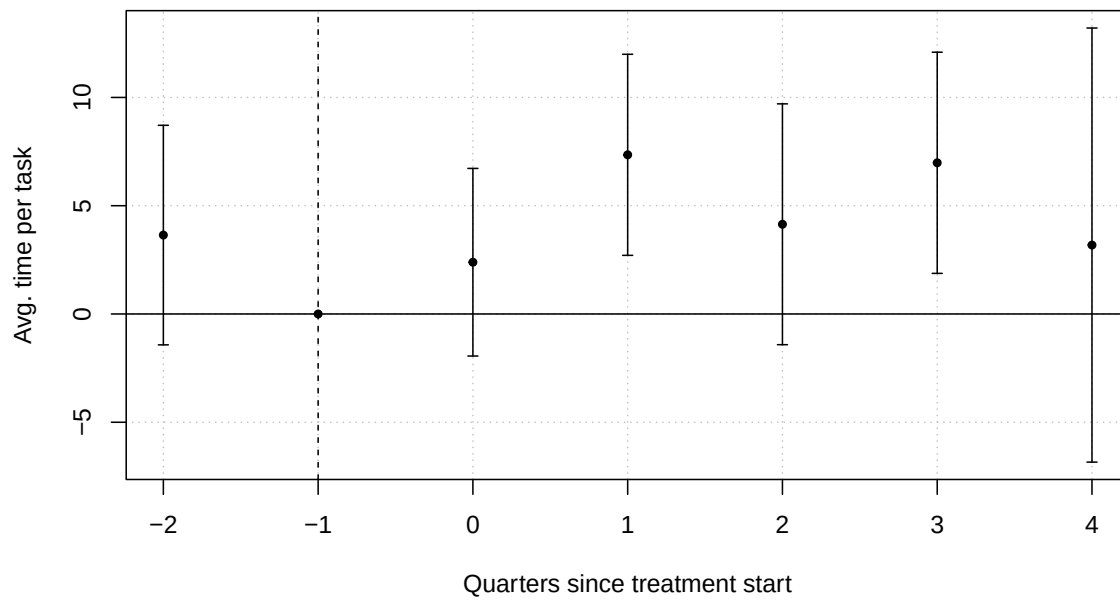


Figure A7: Treatment Effect on Troubleshooting

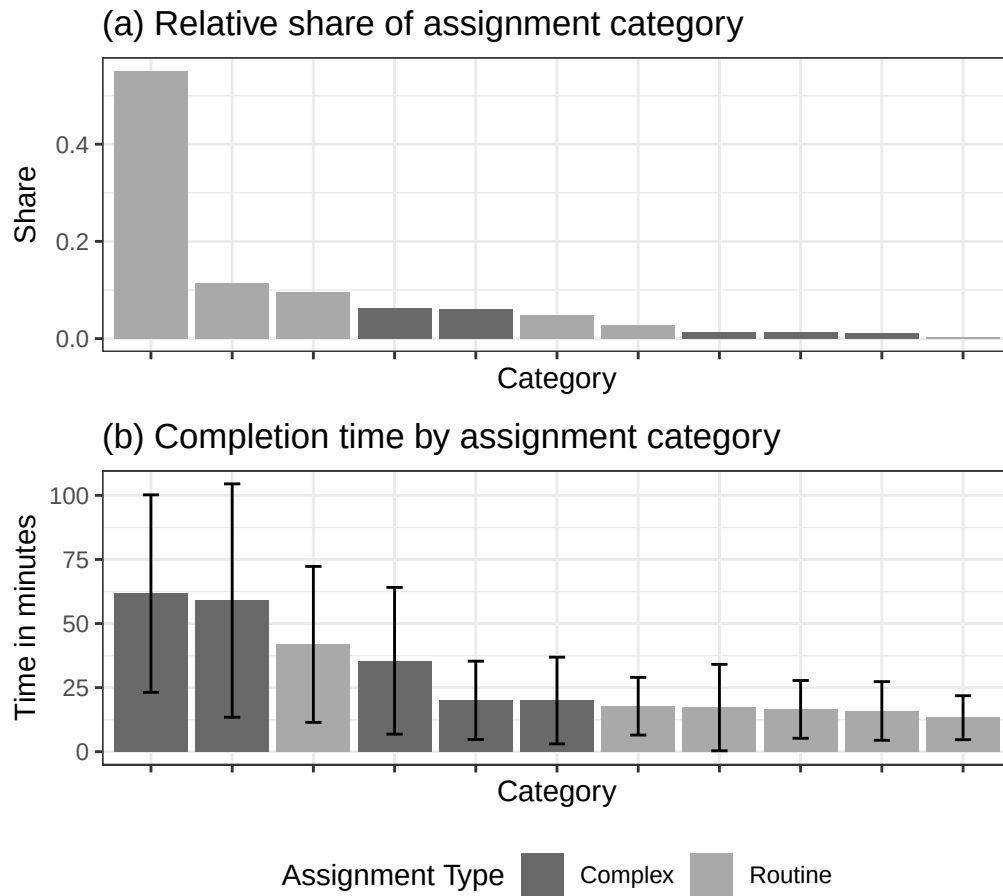


Figure A8: Classification of Work Assignments by Complexity

A.5 Survey Items (Translated)

Item Text	Name
How useful are the following elements of the learning platform to you?	<i>Usefulness</i>
• The skill assessments on the platform • The development ideas you can write down after the skill assessment	
How easy was the handling of the following elements of the platform?	<i>Handling</i>
• The skill assessments on the platform • The development ideas you can write down after the skill assessment	
I participate in the skill assessment because...	
... it is my job.	<i>Obligation</i>
... my manager is expecting it from me.	<i>Demand</i>
... it helps me to find out how I can improve.	<i>Improvement</i>
... to demonstrate to others that I am competent.	<i>Competence</i>
If I do not participate in the skill assessment, it is because...	
... I do not have the time to participate.	<i>Time</i>
... the skill is not relevant for my daily work.	<i>Mismatch</i>
... I already know that I will achieve a high score.	<i>Overconfidence</i>
... I expect to perform poorly.	<i>Incompetence</i>

Item Text	Name
Thinking back to the last 6 months: How often did you have a meeting with your manager to talk about your skill development?	<i>Number of Meetings</i>
Thinking back to the last 6 months, how do you evaluate the following statements regarding your development process?	<i>Supervisor Support</i>
• My manager helps me find the right training measures. • My manager supports me in learning independently. • My manager is interested in what I am currently learning. • My manager helps me try out new solutions, even if I make mistakes in doing so. • Every time I have a good idea for my own development, my manager implements it. • With regards to my development, it often happened that I was presented with accomplished facts by my manager. • I always had a say in the decisions about my own development.	
How satisfied are you with your work at the moment?	<i>Satisfaction</i>
I often think about changing my job.	<i>Turnover</i>
• My manager gives me special recognition when my work performance is especially good. • My manager always gives me positive feedback when I perform well.	<i>Positive Feedback</i>

Item Text	Name
• My manager shows me his/her displeasure when my work is below acceptable standards. • My manager lets me know about it when I perform poorly.	<i>Negative Feedback</i>
Now we ask you to assess what leadership skills you think your manager has.	<i>Leadership</i>
We have included a short description of a person who has this competence.	<i>Skills</i>
• Leadership and coaching: adapts leadership style to the current environment, motivates and mentors others. • Influence on others: is well connected, uses persuasion and their own authority to achieve team goals. • Interpersonal relationships: responds appropriately to the needs of others, understands how to give good feedback, builds trust with others quickly. • Conflict management: anticipates conflicts and complaints and resolves them constructively.	
1. I value the professional skills of my manager.	<i>Perceived</i>
2. I respect my manager's knowledge and expertise.	<i>Competence</i>
3. I am impressed by my manager's knowledge of the job.	